

Sheng Luo PhD ¹

Contact

Email: sheng.luo@duke.edu
Mailing address: 2424 Erwin Rd, Suite 11082, Durham, NC 27705
Telephone: (919) 668-8038 (O)
ORCID: orcid.org/0000-0003-4214-5809

Education

Ph.D. in Biostatistics, Department of Biostatistics, Johns Hopkins University, 2008.
Dissertation: "Mixed Effects Stochastic Process Models of Smoking Cessation Behavior."
Dissertation advisor: Ciprian M. Crainiceanu. Co-advisor: Thomas A. Louis.
M.S. in Statistics/Mathematics, Department of Mathematics, University of Texas at Arlington, 2003.
M.S. in Mechanical Engineering, Huazhong University of Science and Technology, China, 2000.
B.S. in Mechanical Engineering, Huazhong University of Science and Technology, China, 1996.

Professional experience

Full Professor, Department of Biostatistics and Bioinformatics, Duke University, September 2019
- Present

Associate Professor, Department of Biostatistics and Bioinformatics, Duke University, August 16, 2017 - August 2019

Tenured Associate Professor, Department of Biostatistics, School of Public Health, University of Texas Health Science Center at Houston, September 2015 - August 15, 2017

Assistant Professor, Department of Biostatistics, School of Public Health, University of Texas Health Science Center at Houston, July 2008-August 2015

Predoctoral fellow and statistical consultant, Biostatistics Branch, Division of Cancer Epidemiology and Genetics, National Cancer Institute, National Institutes of Health, 2005-2008, Mentor: Dr. Nilanjan Chatterjee

Research assistant, Department of Molecular Microbiology and Immunology, Johns Hopkins University, 2004-2005

Research assistant, Department of Environmental Health Sciences, Johns Hopkins University, 2004-2005

Research assistant, Department of Mathematics, University of Texas at Arlington, 2000-2003

¹Last updated March 24, 2026

Honors and awards

1. **Fellow, American Statistical Association (2020)**, citation reads: "For excellent contributions to statistical methods in longitudinal and survival analysis; for broad scientific contributions to the field of neurological diseases; for exemplary mentorship of students and significant service to the profession. "
2. **Elected Member, International Statistical Institute (2020)**
3. Young Investigator Award, the American Statistical Association, Statistics in Epidemiology Section, 2007
4. Laha Award, the Institute of Mathematical Statistics, 2007
5. Fellow Award for Research Excellence, the National Institutes of Health (NIH), 2007
6. Distinguished Student Paper Award, the International Biometrics Society, Eastern North American Region (ENAR), 2007
7. Overall first-place winner of the Scientific Poster Competition, the Delta Omega Public Health Honor Society, Alpha Chapter, Johns Hopkins Bloomberg School of Public Health, 2007
8. Louis I. and Thomas D. Dublin Award for the Advancement of Epidemiology and Biostatistics, Johns Hopkins Bloomberg School of Public Health, 2007
9. Cancer Training Research Award, the National Cancer Institute, NIH, 2005–2008

Grants and contract awards

Research Support

Ongoing research support

1. NIH/NIA, R01AG064803, "Integrative modeling and dynamic prediction of Alzheimer's disease", 9/15/2020 - 5/31/2026, role: **PI**.
2. NIH/NIA, P30AG072958, "Duke/UNC Alzheimer's Disease Research Center (ADRC)", 9/1/2021-6/30/2026, MPIs: Heather Whitson, Gwenn Garden (UNC), role: **Co-PI** of Data Management and Statistics Core.
3. NIH/NIDDK, U01DK140923, "The Bidirectional Gut-Microbiota-Brain Axis in Parkinson's disease: integrating mechanistic biomarkers of disease severity and progression", 8/15/2024-6/30/2029, MPIs: Ali Keshavarzian (Rush), Christopher G. Goetz (Rush), and Amol Sharma (Augusta), role: **PI** of Duke sub-contract.
4. NIH/NCI, R01CA297227, "AI for Differentiation of Low vs High Risk Thyroid Nodules on Ultrasound", 9/15/2025-8/31/2029, MPIs: Ben Wildman-Tobriner, Maciej Mazurowski, Miriam Morey, role: co-investigator.
5. NIH/NCBIB, P41EB028744, "Center for Virtual Imaging Trials", 4/1/2021-12/31/2026, PI: Ehsan Samei, role: co-investigator.
6. NIH/NIA, R01AG080048, "Common mechanistic biomarkers of vascular and neuro-degeneration", 1/1/2023-12/31/2027, PI: Yongmei Liu, role: co-investigator.
7. NIH/NIA, R01AG081322, "Metabolic age to define influences of the lipidome on brain aging in Alzheimer's disease", 5/15/2023-3/31/2028, MPI: Rima Kaddurah-Daouk, Matthias Arnold, Peter Meikle, role: co-investigator.
8. NIH/NIA, R01AG084098, "Automated high-purity exosome isolation-based AD diagnostics system (AHEADx)", 9/1/2023-5/31/2028, PI: Tony Jun Huang, role: co-investigator.

9. NIH/NIA, R01AG082335, "Biopsy-based single cell assays for olfactory sensory changes in an Alzheimer's disease cohort", 2/1/2024-11/30/2028, PI: Bradley Goldstein, role: co-investigator.
10. NIH/NCI, R01CA261457, "Computer-aided triage of body CT scans with deep learning", 8/16/2023-7/31/2027, PI: Joseph Lo, role: co-investigator.
11. NIH/NHLBI, R01HL155293, "Accuracy and precision in CT quantification of COPD through virtual imaging trials", 7/1/2021-6/30/2026, PI: Ehsan Abadi, role: co-investigator.
12. NIH/NIBIB, R01EB031575, "Harmonization of breast MRI data", 9/15/2022-6/30/2026, PI: Maciej J Mazurowski, role: co-investigator.
13. NIH, U19NS115388, "Determinants of Incident Stroke Cognitive Outcomes and Vascular Effects on Recovery", 9/19/2019 - 8/31/2026, PI: Rebecca Gottesman (John Hopkins University), MD, role: co-investigator.

Completed research support

14. NIH/NIA, R56AG076622, "Trajectories of blood-based biomarkers of AD, their determinants, and ability to predict cognitive impairment", 9/15/2022-8/31/2024, MPIs: Yongmei Liu, Sheng Luo, role: **MPI**.
15. Parkinson's Foundation, award ID: 858345, "Parkinson's Outcomes Project - Statistical Analysis Core", 3/1/2021- 10/31/2023, role: **PI**.
16. CHDI Foundation, project ID: A-16209, "Novel latent trait models of Huntington's disease", 7/1/2021- 12/31/2022, role: **PI**.
17. NIH/NINDS, R01NS091307, "Statistical methods for clinical trials with multivariate longitudinal outcomes", 9/30/2015 - 12/31/2020, role: **PI**.
18. NIH/NIA, R56AG064803, "Integrative modeling and dynamic prediction of Alzheimer's disease", 9/1/2019 - 8/31/2020, role: **PI**.
19. NIH/NIA, R56AG062302, "Shared genetic, epigenetic, and transcriptomic profiles between AD and PTSD: molecular insights into the heterogeneity of neuropsychiatric symptoms in Alzheimers Disease", 9/30/2018 - 8/31/2020, role: **co-PI**.
20. NIH/NICHD, 1R01HD096331, "Vestibulodynia: Understanding Pathophysiology and Determining Appropriate Treatments (Vestibulodynia: UPDATE)", 9/11/2018 - 5/31/2023, PI: Andrea Nackley, PhD, role: co-investigator.
21. NIH/NICHD, 1P50HD093074, "Co-occurring ADHD in young children with ASD: Precursors, detection, neural signatures, and early treatment", 9/7/2017 - 7/31/2022, PI: Geraldine Dawson, PhD, role: faculty statistician.
22. CHDI Foundation, "Dynamic monitoring and risk prediction of Huntington's disease", 7/1/2019-6/30/2020, role: **PI**.
23. NIH/NIDA, R34DA050267, "HEAL Consortium: Establishing innovative approaches for the Healthy Brain and Child Development Study", 9/30/2019 - 3/31/2021, PI: Brian Smith, MD, role: co-investigator.
24. AKili, Inc, "Software Treatment for Actively Reducing Severity of ADHD as Adjunctive Treatment to Stimulant (STARS-ADHD-Adjunctive)", PI: Daniel Laskowitz, MD, role: faculty statistician.
25. KemPharm, Inc, "A multicenter, dose-optimized, double-blind, randomized, placebo-controlled, parallel efficacy laboratory classroom study with KP415 in children with attention-deficit/hyperactivity disorder (ADHD)", PI: Scott H. Kollins, PhD, role: faculty statistician.
26. NIH/NICHD, 5U01HD073984, "Study of oxytocin in autism to improve reciprocal social behaviors (SOARS-B)", 9/4/2012 - 5/31/2028, PI: Linmarie Sikich, MD, role: faculty statistician.

27. NIH/NIMH, "A phase 2a study to evaluate the Kappa Opioid Receptor as a target for the treatment of mood and anxiety spectrum disorders by evaluation of whether LY2456302 engages key neural circuitry related to the hedonic response (FAST-MAS KOR 2a)", PI: Andrew D. Krystal, MD, role: faculty statistician.
28. Intra-Cellular Therapies, Inc, "A randomized, placebo-controlled, double-blind study of safety, tolerability, pharmacokinetics and pharmacodynamics of multiple doses of ITI-214 in patients with idiopathic Parkinson's disease", PI: Laurie Sanders, PhD, role: faculty statistician.
29. CHDI Foundation (a privately-funded, not-for-profit biomedical research organization devoted to Huntington's disease), "FuRST 2.0 (Functional Rating Scale Taskforce) Development and Testing", 15% effort, project period: 2/1/2015-12/31/2017, role: **PI**.
30. International Parkinson and Movement Disorder Society (a professional society representing Parkinson's disease, related neurodegenerative and neurodevelopmental disorders, hyperkinetic movement disorders, and abnormalities in muscle tone and motor control), "Validating the Uniform Parkinson's Disease Rating Scale", 12/15/2012 - 11/30/2017, role: **PI**.
31. International Parkinson and Movement Disorder Society, "Missing values and differential item functioning analysis of the Unified Dyskinesia Rating Scale (UDysRS)", 1/1/2017 - 12/31/2017, role: **PI**.
32. NIH/NINDS, 5U01NS043127, "Parkinson's Disease Clinical Trial: Statistical Center", 10% effort, 06/01/2016 - 05/31/2017, role: **PI**, took over from Dr. Barbara Tilley since June 1, 2016.
33. NIH/NCATS, 5KL2TR000370, "Innovative Statistical Methods for Analyzing Parkinson's Disease Study Data", 75% effort (6/1/2013 - 9/29/2015), 50% (9/30/2015 - 5/31/2016), role: **PI**.
34. National Parkinson Foundation, "Determining the Effects of Early Exercise and Physical Therapy on Outcomes Using the Parkinson's Outcomes Project Dataset", project period: 6/15/2015-6/14/2016, role: **PI**.
35. International Parkinson and Movement Disorder Society, "The use of the MDS-UPDRS validation and translation datasets to investigate DIF and missing values", 3/1/2014 - 2/28/2015, role: **PI**.
36. Dystonia Coalition/Office of Rare Diseases Research (ORDR)/NINDS, "Item Response Theory: An Individualized Approach to Evaluating a Composite Cervical Dystonia Rating Scale", 7/9/2012-12/31/2013, role: consultant.
37. NIH-NINDS, 5U01NS043127, P.I. Barbara Tilley, "Parkinson's Disease Clinical Trial: Statistical Center", 40% effort, 09/01/2009 - 05/31/2013, role: Co-Investigator.
38. Subcontracted from Baylor College of Medicine, through a NIH P30 core grant, PI: Janet Butel, "Baylor-UTHouston Center for AIDS Research", 5% effort, 8/10/2010 - 5/31/2013, role: Co-Investigator.
39. NIH-NHLBI, RC2HL102419-0110, PI: Eric Boerwinkle, "Building on GWAS for NHLBI-diseases: the CHARGE consortium", 10% effort, 11/17/2009-7/31/2012, role: Co-Investigator.
40. UTHSC-SPH, PRIME award, "Classifying Ipsilateral Breast Tumor Relapse: a Novel Bayesian Approach", 09/01/2009 - 08/31/2010, role: **PI**.
41. NIH-NEI, supplement to 5U10EY012471, PI: Robert Hardy, "A Hybrid Method in Combining Treatment Effects from Matched and Unmatched Studies", 25% effort, 10/1/2009 - 9/29/2010, role: Co-Investigator.

Professional Activities

Service to Professional Society/Foundation

1. International Parkinson and Movement Disorder Society (MDS), Clinical Outcome Assessments (COA) Translation Steering Committee, **Chair**, Oct 1, 2021 - present. Duties: Recommend and approve non-English translations of the MDS-owned scales MDS-UPDRS, MDS-NMS, and UDysRS; Assist in the development of non-English translation teams, monitor translation development and oversee all aspects of the statistical analysis at each phase of development; Make recommendations for best practices in the translation and validation of digital tools acquired by the Society; Manage use of generated rating scales data owned by MDS.
2. National Vaccine Injury Compensation Program (VICP), Health Resources and Services Administration (HRSA), Department of Health and Human Services (HHS), **Medical Panel Expert**, 2021 - present. Duties: review medical records and relevant scientific literature, prepare written reports, and serve as government expert witnesses in hearings before the Court.
3. National Alzheimer's Coordinating Center (NACC), **Data Core Steering Committee**, Fall 2022 - Fall 2025. 2-year term as member and 1-year term as Chair.
4. National Parkinson Study Group (PSG), **Scientific Review Committee**, Nov 1, 2022 - Oct 31, 2025, 3-year term.
5. Huntington Study Group (HSG), **Scientific Review Committee**, Jan, 2023 - present.
6. IQVIA Data Safety Monitoring Board, 2021 - present.
7. National Institute on Aging (NIA) **Biobank Scientific Review Committee**, Oct 2020 - present.
8. American Statistical Association, Biometrics Section Chair (January 2019 - December 2019).
9. American Statistical Association, Committee on Career Development, January 2018 - December 2019.
10. MDS Rating Scales Translation Committee, Chair, June 30, 2017 - September 30, 2021
11. The International Parkinson and Movement Disorder Society (MDS) Rating Scales Steering Committee, vice-chair (September 26, 2019 - September 30, 2021); member (June 30, 2017 - September 25, 2019)
12. MDS Scales Development Committee, June 15, 2013 - June 30, 2017
13. The Parkinson Study Group (PSG) Scientific Review Committee (SRC), December 2013 - January 2017
14. Eastern North American Region (ENAR) of the International Biometric Society (IBS) Regional Advisory Board (RAB), January 1, 2015 - December 31, 2017.
15. International Chinese Statistical Association (ICSA), Biometrics Section Chair, January 2016-December 2016.
16. American Statistical Association, Biometrics Section Publication Officer, January 2015 - December 2016.

Grant Review Services

1. NIH reviewer on Special Emphasis Panel/Scientific Review Group ZRG1 HSS-R (90)S. Review Date: Oct 9-10, 2025, SRO: Rhoda Keese Moise.

2. NIH reviewer on Special Emphasis Panel /Scientific Review Group ZAG1-Zu-5(M2)B. Review Date: Apr 15-16, 2025, SRO: Janetta Lun.
3. **NIH NMBH (Neurological, Mental and Behavioral Health) Study Section Chartered Member, 07/01/2022-06/30/2024.** Grant reviewed on 10/17 - 10/18/2022, 2/27/2023-2/28/2023, 6/29/2023-6/30/2023, 2/21/2024-2/22/2024, 5/30/2024-5/31/2024.
4. **NIH NAME (Neurological, Aging and Musculoskeletal Epidemiology) Study Section Chartered Member, 07/01/2020-06/30/2022.** Grant reviewed on 10/14 - 10/16/2020; 02/24 - 02/26/2021; 07/07 - 07/09/2021, 10/06 - 10/07/2021, 02/09 - 02/10/2022, 06/08 - 06/09/2022.
5. The Parkinson Study Group (PSG) grant review (review Date: 05/05/2014, 05/22/2015, 07/07/2015, 05/2015, 11/2015, 07/22/2016, 01/27/2017)
6. NIH reviewer on Special Emphasis Panel /Scientific Review Group 2015/10 ZHL1 CSR-D (O1). Review Date: July 29, 2015, SRO: Keary Cope.
7. NIH reviewer on National Heart, Lung, And Blood Institute Clinical Trials Review Committee, CLTR-MA, Review Date: March 3-4, 2016, SRO: Keary Cope.
8. NIH reviewer on Risk Prevention and Health Behavior Special Emphasis Panel, ZRG1-RPHB-C-11. Review Date: November 17-18, 2016, SRO: Claire Gutkin.
9. NIH reviewer on NINDS Special Emphasis Panel, ZNS1-SRB-C-04. Review Date: November 28, 2016, SRO: Tiziana Cogliati.
10. NIH reviewer on Healthcare Delivery and Methodologies (HDM-11) Small Business: Health Informatics Study Section, ZRG1-HDM-G-11. Review Date: March 13, 2017, SRO: Yvonne Ferguson.
11. NIH reviewer on Healthcare Delivery and Methodologies (HDM-11) Small Business: Health Informatics Study Section, ZRG1-HDM-A-11. Review Date: June 26, 2017, SRO: Michael Bloom.
12. NIH reviewer on NINDS Clinical Trial Readiness meeting. Review Date: July 14, 2017, SRO: Ana Olariu.
13. NIH reviewer on Small Business: Health Informatics Special Emphasis Panel, ZRG1-HDM-Z-11. Review Date: November 03, 2017, SRO: Peter Kozel.
14. NIH reviewer on Small Business: Health Informatics Special Emphasis Panel, ZRG1-HDM-H-11. Review Date: March 26-27, 2018, SRO: Sudha Veeraghavan.
15. NIH reviewer on Biobehavioral Mechanisms of Emotion, Stress and Health Study Section, MESH, Review Date: June 7-8, 2018, SRO: Maribeth Champoux.
16. NIH reviewer on Health Decisions for Older People Study Section, National Institute of Aging, Review Date: September 12, 2018, SRO: Carmen Moten.
17. NIH reviewer on Neurological, Aging and Musculoskeletal Epidemiology (NAME) Study Section, Review Date: October 11-12, 2018, SRO: Heidi B. Friedman.
18. NIH reviewer on NIH/CSR Anonymization Study (HDM), Review period: March 2019, SRO: Jeremy Braithwaite.
19. NIH reviewer on Special Emphasis Panel, ZRG1 PSE-A (56) R, Review date: July 18-19, 2019, SRO: Delia Olufokunbi Sam.

20. NIH reviewer on Neurological, Aging and Musculoskeletal Epidemiology (NAME) Study Section, Review Date: October 17-18, 2019, SRO: Heidi B. Friedman.
21. NIH reviewer on Special Emphasis Panel, ZAG1ZIJ-G(M3), Review Date: Feb 25, 2022, SRO: Maurizio Grimaldi.

Data and Safety Monitoring Board Services

1. SEGA - SEdation versus General Anesthesia for Endovascular Therapy in Acute Ischemic Stroke a Randomized Comparative Effectiveness Trial. ClinicalTrials.gov Identifier: NCT03263117, PIs: Drs. Peng Roc Chen, Andrew Barreto, Carlos A. Artime, McGovern Medical School at The University of Texas Health Science Center at Houston, status: completed.
2. An Open-Label Phase I/II Dose Finding Study to Evaluate the Safety, Tolerability, and Preliminary Efficacy of Striatal Administration of BV-101 in Adult Subjects with Early Manifest Huntington's Disease (HD). Regulatory Agency Identifier Number: EudraCT: 2021-003052-16. Sponsor: Asklepios BioPharmaceutical. Status: ongoing.

Steering Committee Services for Clinical Trials

1. A Phase 2, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy, Safety, Tolerability, Pharmacodynamics, and Pharmacokinetics of BIA 28-6156 in Subjects With Parkinson's Disease With a Pathogenic Variant in the Glucocerebrosidase (GBA1) Gene. Sponsor: BIAL R& D Investments S.A. Status: ongoing.

Editorial Boards

1. Associate Editor, BMC Medical Research Methodology, February 2016 - present, manuscripts handled (22).
2. Associate Editor, Journal of Alzheimer's Disease, January 2018 - present.
3. Editorial board member, Cancer Medicine, June 2018 - present.

Journal Referee (Number in parenthesis indicates # of manuscripts reviewed): American Journal of Epidemiology (4), Alzheimer's & Dementia (3), Annals of Applied Statistics (3), Annals of Epidemiology (1), Annals of Human Genetics (1), Assessment (1), Australian & New Zealand Journal of Statistics (1), Bayesian Analysis (3), Biometrical Journal (1), Biometrics (14), Biostatistics (7), BMC Health Services Research (1), BMC Medical Informatics and Decision Making (1), BMC Medical Research Methodology (6), BMC Medicine (1), BMC Neurology (1), BMJ Open (1), Canadian Journal of Statistics (1), Cancer Cell (6), Computational Statistics & Data Analysis (6), Computer Methods and Programs in Biomedicine (1), Communications in Statistics-Theory and Methods (1), Computer Methods and Programs in Biomedicine (1), Journal of Alzheimers Disease (2), Journal of American Statistical Association (6), Journal of Applied Probability and Statistics (1), Journal of Applied Statistics (5), Journal of Clinical Oncology (8), Journal of Consulting and Clinical Psychology (4), Journal of the Royal Statistical Society-Series C: Applied Statistics (2), Journal of Probability and Statistics (1), Journal of Statistical Computation and Simulation (1), Metron (1), Movement Disorders (13), Movement Disorders Clinical Practice (1), Nature (1), Nature Biotechnology (4), Nature Communications (9), Nature Chemical Biology (1), Nature International Journal of Obesity (1), Nature Medicine (5), Plos One (2), Psychology, Health & Medicine (1), Scientific Reports (1), Statistics in Biopharmaceutical Research (1), Statistics in Medicine (17), Statistical Methods in Medical Research (5), WIREs

Publications

* represents student/postdoc author under Sheng Luo's supervision; ** represents student author of which Sheng Luo serves as committee member; † represents Sheng Luo as corresponding author.

Statistical methodology papers

1. **Luo S**[†], Crainiceanu CM, Louis TA, Chatterjee N. (2008). Analysis of smoking cessation patterns using a stochastic mixed-effects model with a latent cured state. *Journal of The American Statistical Association*, 103(483), 1002-13, PMID: 19305513, PMCID: PMC2658598.
2. **Luo S**[†], Crainiceanu CM, Louis TA, Chatterjee N. (2009). Bayesian inference for smoking cessation with a latent cure state. *Biometrics*, 65(3), 970-8, PMID: 19173701, PMCID: PMC3856570.
3. **Luo S**[†], Mukherjee B, Chen J, Chatterjee N. (2009). Shrinkage estimation for robust and efficient screening of single SNP association from case-control genome-wide association studies. *Genetic Epidemiology*, 33(8), 740-50, PMID: 19434716, PMCID: PMC3103068.
4. Chatterjee N, Chen YH, **Luo S**, Carroll RJ. (2009). Analysis of case-control association study: SNPs, imputation and haplotypes, *Statistical Science*, 24(4), 489-502, PMID: 20543902, PMCID: PMC2883271.
5. Crainiceanu CM, Caffo B, **Luo S**, Zippunikov V, Punjabi NM. (2011). Population value decomposition, a framework for the analysis of image populations. *Journal of the American Statistical Association*, discussion paper, 106(495), 775-90, PMID: 24415813, PMCID: PMC3886284.
6. Crainiceanu CM, Caffo B, **Luo S**, Zippunikov V, Punjabi NM. (2011). Rejoinder to comments on "Population value decomposition, a framework for the analysis of image populations". *Journal of the American Statistical Association*, 106(495), 803-6.
7. **Luo S**[†], Yi M*, Huang X, Hunt KK. (2013). A Bayesian model for misclassified binary outcomes and correlated survival data with applications to breast cancer. *Statistics in Medicine*, 32(13), 2320-34, PMID: 22996169, PMCID: PMC3897718.
8. **Luo S**[†], Ma J*, Kiebertz KD. (2013). Robust Bayesian inference for multivariate longitudinal data using normal/independent distributions, *Statistics in Medicine*, 32(22), 3812-28, PMID: 23494809, PMCID: PMC3884581.
9. Chen Y, **Luo S**, Chu H, Wei P. (2013). Bayesian inference on risk differences: an application to multivariate meta-analysis of adverse events in clinical trials, *Statistics in Biopharmaceutical Research*, 5(2), 142-50, PMID: 23853700, PMCID: PMC3706106.
10. Byun J**, Lai D, **Luo, S**, Risser J, Tung B, Hardy RJ. (2013). A hybrid method in combining treatment effects from matched and unmatched studies, *Statistics in Medicine*, 32(28), 4924-37, PMID: 23839782, PMCID: PMC3887129.
11. **Luo S**[†]. (2013). Joint analysis of stochastic processes with application to smoking patterns and insomnia, *Statistics in Medicine*, 32(29), 5133-44, PMID: 23913574, PMCID: PMC3856619.
12. **Luo S**[†], Chen Y, Su X*, Chu H. (2014). *mmeta*: An R package for multivariate meta-analysis, *Journal of Statistical Software*, 56(11), 1-26, PMID: 24904241, PMCID: PMC4043353.
13. **Luo S**[†]. (2014). A Bayesian approach to joint analysis of multivariate longitudinal data and parametric accelerated failure time, *Statistics in Medicine*, 33(4), 580-94, PMID: 24009073, PMCID: PMC3947121.

14. **Luo S[†]**, Su X*, DeSantis S, Huang X, Yi M*, Hunt KK. (2014). Joint model for a diagnostic test without a gold standard in the presence of a dependent terminal event, *Statistics in Medicine*, 33(15), 2554-66, PMID: 24473943, PMCID: PMC4209250.
15. Chen Y, **Luo S**, Chu H, Su X*, Nie L. (2014). An empirical Bayesian method for multivariate meta-analysis with an application in clinical trials. *Communication in Statistics-Theory and Methods*, 43(16), 3536-51, PMID: 25089070, PMCID: PMC4115294.
16. **Luo S[†]**, Wang J*. (2014). Bayesian hierarchical model for multiple repeated measures and survival Data: an application to Parkinson's disease, *Statistics in Medicine*, 33(24), 4279-91, PMID: 24935619, PMCID: PMC4184935.
17. **Luo S[†]**, Su X*, Yi M*, Hunt KK. (2015). Simultaneous inference of a misclassified outcome and competing risks failure time data, *Journal of Applied Statistics*, 42(5), 1080-90, PMID: 25821331, PMCID: PMC4372162.
18. Chen Y, Chu H, **Luo S**, Nie L, Chen S. (2015). Bayesian analysis on meta-analysis of case-control studies accounting for within study correlation. *Statistical Methods in Medical Research*, 24(6), 836-55, PMID: 22143403, PMCID: PMC3683108.
19. Wang J*, **Luo S[†]**. (2016). Augmented Beta rectangular regression models: A Bayesian perspective, *Biometrical Journal*, 58(1), 206-21, PMID: 26289406, PMCID: PMC5064841 [**** An earlier version won Jue Wang a Student Travel Award from Parkinson's Disease Foundation in year 2014. ****]
20. **Luo S[†]**, Chan W, Detry M**, Massman PJ, Doody RS. (2016). Binomial regression with a misclassified covariate and outcome, *Statistical Methods in Medical Research*, 25(1), 101-17, PMID: 22421539, PMCID: PMC3883897.
21. **Luo S[†]**, Lawson AB, He B*, Elm J, Tilley BC. (2016). Bayesian multiple imputation for missing multivariate longitudinal data from a Parkinson's disease clinical trial, *Statistical Methods in Medical Research*, 25(2), 821-37, PMID: 23242384, PMCID: PMC3883900.
22. Benoit J**, Chan W, **Luo S**, Yeh HW, Doody R. (2016). A hidden Markov model approach to analyze longitudinal ternary outcomes when some observed states are possibly misclassified, *Statistics in Medicine*, 35(9), 1549-57, PMID: 26782946, PMCID: PMC4821697.
23. Chen G*, **Luo S[†]**. (2016). Robust Bayesian hierarchical model using normal/independent distributions, *Biometrical Journal*, 58(4), 831-51, PMID: 26711558, PMCID: PMC5064853.
24. He B*, **Luo S[†]**. (2016). Joint modeling of multivariate longitudinal measurements and survival data with applications to Parkinson's disease, *Statistical Methods in Medical Research*, 25(4), 1346-58, PMID: 23592717, PMCID: PMC3883896.
25. Li L, **Luo S**, Hu B, Greene T. (2017). Dynamic prediction of renal failure using longitudinal biomarkers in a cohort study of chronic kidney disease, *Statistics in Biosciences*, 9(2), 357-78, PMID: 29250207, PMCID: PMC5726783.
26. Wang J*, **Luo S[†]**. (2017). Bayesian multivariate augmented Beta rectangular regression models for patient-reported outcomes and survival data, *Statistical Methods in Medical Research*, 26(4), 1684-99, PMID: 26037528, PMCID: PMC4457342.
27. Zhu H*, **Luo S**, DeSantis, S. (2017). Zero-inflated count models for longitudinal measurements with heterogeneous random effects, *Statistical Methods in Medical Research*, 26(4), 1774-86, PMID: 26113383, PMCID: PMC5045322.
28. Su X*, **Luo S[†]**. (2017). Analysis of censored longitudinal data with skewness and dependent censoring, *Communications in Statistics-Simulation and Computation*, 46(7), 5378-91, PMID: 29056818, PMCID: PMC5646848.

29. Lin L*, Luo S[†], Chen BE, Davis B. (2017). Bayesian analysis of multi-type recurrent events and dependent termination with nonparametric covariate function, *Statistical Methods in Medical Research*, 26(6), 2869-84, PMID: 26546256, PMCID: PMC5061632.
30. Wang J*, Luo S[†]. (2017). Multidimensional latent trait linear mixed model: an application in clinical studies with multivariate longitudinal outcomes, *Statistics in Medicine*, 36(20), 3244-56, PMID: 28569393, PMCID: PMC5540878.
31. Wang J*, Luo S[†], Li L. (2017). Dynamic prediction for multiple repeated measures and event time data: an application to Parkinson's disease, *Annals of Applied Statistics*, 11(3), 1787-809, PMID: 29081873, PMCID: PMC5656296, **[** An earlier version won Jue Wang ENAR Distinguished Student Paper Award in year 2016. **]**
32. Li K*, Luo S[†]. (2017). Functional joint model for longitudinal and time-to-event data: an application to Alzheimer's disease, *Statistics in Medicine*, 36(22), 3560-72, PMID: 28664662, PMCID: PMC5583028.
33. Chen G*, Luo S[†]. (2018). Bayesian hierarchical joint modeling using skew-normal/independent distributions, *Communications in Statistics-Simulation and Computation*, 47(5):1420-38, PMID: 30174369; PMCID: PMC6114938.
34. Zhu H*, DeSantis SM, Luo S. (2018). Joint modeling of longitudinal zero-inflated count and time-to-event data: A Bayesian perspective, *Statistical Methods in Medical Research*, 27(4), 1258-70, PMID: 27460540, PMCID: PMC5269555.
35. Lin L*, Luo S[†], Davis B. (2018). Bayesian regression model for recurrent event data with event-varying covariate effects and event effect, *Journal of Applied Statistics*, 45(7), 1260-76, PMID: 29755162; PMCID: PMC5945197.
36. Yang M*, Luo S[†], DeSantis SM. (2019). Bayesian quantile regression joint models: inference and dynamic predictions, *Statistical Methods in Medical Research*, 28(8), 2524-37, PMID: 29962288, PMCID: PMC6050160.
37. Wang J*, Luo S[†], (2019). Joint modeling of multiple repeated measures and survival data from multidimensional latent trait linear mixed model, *Statistical Methods in Medical Research*, 28(10-11), 3392-3403, PMID: 30306833, PMCID: PMC6478574.
38. Li K*, Luo S[†]. (2019). Bayesian functional joint models for multivariate longitudinal and time-to-event data, *Computational Statistics & Data Analysis*, 129, 14-29, PMID: 30559575, PMCID: PMC6294314.
39. Li K*, Luo S[†]. (2019). Dynamic predictions in Bayesian functional joint models for longitudinal and time-to-event data: an application to Alzheimer's disease, *Statistical Methods in Medical Research*, 28(2), 327-42, PMID: 28750578, PMCID: PMC5557714.
40. Li K*, Luo S, Yuan S, Mt-Isa S. (2019). A Bayesian approach for individual-level drug benefit-risk assessment, *Statistics in Medicine*, 38(16), 3040-52, PMID: 30989691, PMCID: PMC6681911, **[** An earlier version won Kan Li Honorable Mention Award in the Biopharmaceutical Section's Student Paper Competition in year 2018. **]**.
41. Li K*, Luo S[†]. (2019). Dynamic predictions of Alzheimer's disease progression using features of multiple longitudinal outcomes and time-to-event data, *Statistics in Medicine*, 38(24), 4804-18, PMID: 31386218, PMCID: PMC6800781.
42. Halabi S, Li C*, Luo S. (2019). Developing and validating risk assessment models of clinical outcomes in modern oncology, *Journal of Clinical Oncology: Precision Oncology*, 3, 1-12, PMID: 31840130, PMCID: PMC6908945.
43. Li C*, Xiao L, Luo S. (2020). Fast covariance estimation for multivariate sparse functional data, *Stat*, 9(1), e245, PMID: 34262756, PMCID: PMC8276768.

44. Ren X*, Wang J, **Luo S[†]**. (2021). Dynamic prediction using joint models of longitudinal and recurrent event data: A Bayesian perspective, *Biostatistics & Epidemiology*, 5(2), 250-266, PMID: 34926969, PMCID: PMC8673593.
45. Lin J*, Li K, **Luo S[†]**. (2021). Functional survival forests for multivariate longitudinal outcomes: dynamic prediction of Alzheimer's disease progression, *Statistical Methods in Medical Research*, 30(1), 99-111, PMID: 32726189, PMCID: PMC7855476.
46. Zou H*, Li K, Zeng D, **Luo S[†]**. (2021). Bayesian inference and dynamic prediction of multivariate joint model with functional data: an application to Alzheimer's disease, *Statistics in Medicine*, 40(30):6855-6872, PMID: 34649301, PMCID: PMC8671252.
47. Lin J*, **Luo S[†]**. (2022). Deep learning for the dynamic prediction of multivariate longitudinal and survival data, *Statistics in Medicine*, 41(15):2894-907, PMID: 35347750, PMCID: PMC9232978.
48. Li C*, Xiao L, **Luo S**. (2022). Joint Model for Survival and Multivariate Sparse Functional Data with Application to a Study of Alzheimer's Disease, *Biometrics*, 78:435-447, PMID: 33501651, PMCID: PMC8310894.
49. Zou H*, Xiao L, Zeng D, **Luo S[†]**. (2023). Multivariate functional mixed model with MRI data: an application to Alzheimer's Disease, *Statistics in Medicine*, 42(10):1492-1511, PMID:36805635, PMCID: PMC10133011.
50. Zou H*, Zeng D, Xiao L, **Luo S[†]**. (2023). Bayesian inference and dynamic prediction for multivariate longitudinal and survival data, *Annals of Applied Statistics*, 17(3):2574-2595, PMID: 37719893, PMCID: PMC10500582.
51. Koner S*, **Luo S[†]**. (2024). Projection-based two-sample inference for sparsely observed multivariate functional data, *Biostatistics*, 25(4):1156-1177, PMID: 38413051, PMCID: PMC11639128.
52. Zou H*, Xiao L, Zeng D, **Luo S[†]**. (2025). Dynamic prediction with multivariate longitudinal outcomes and longitudinal magnetic resonance imaging data, *Annals of Applied Statistics*, 19(1):505-528, PMID: 40584459, PMCID: PMC12206078.
53. Koner S*, **Luo S[†]**. (2025). Power and sample size calculation of two-sample projection-based testing for sparsely observed functional data, *Stat*, 14(1):e70046, PMID: 41536944, PMCID: PMC12799244.
54. Xu X*, Ghosh D*, **Luo S[†]**. (2025). A novel longitudinal rank-sum test for multiple primary endpoints in clinical trials: Applications to neurodegenerative disorders, *Statistics in Biopharmaceutical Research*, 17(4):616-626, PMID: 40857536, PMCID: PMC12352412.
55. Ghosh D*, **Luo S[†]**. (2025). A non-parametric U-statistic testing approach for multi-arm clinical trials with multivariate longitudinal data, *Journal of Multivariate Analysis*, 209:105447, PMID: 40416930, PMCID: PMC12097542.
56. Guo Y*, Zou H*, Alam MS*, **Luo S[†]**. (2025). Integrative multi-omics and multivariate longitudinal data analysis for dynamic prediction in Alzheimer's disease, *Statistics in Medicine*, 44:e70105, PMID: 40387018, PMCID: PMC12092054.
57. Ghosh D*, Xu X*, **Luo S[†]**. (2025). Power and sample size calculation for multivariate longitudinal trials using the longitudinal rank sum test, *Statistics in Medicine*, 44:e70261, PMID: 40947402, PMCID: PMC12614807.
58. Alam MS*, Choi D*, Koner S*, **Luo S[†]**. (2025). Dynamic prediction using functional latent trait joint models for multivariate longitudinal outcomes: An application to Parkinson's disease, *Statistics in Medicine*, 44(23-24):e70285, PMID: 41103042, PMCID: PMC12614809.
59. Lin J*, **Luo S[†]**. (2026). Deep learning for the joint analysis of item-level longitudinal and survival data, *Journal of Applied Statistics*, published online on Feb 9, 2026, doi.org/10.1080/02664763.2026.2625122.

60. Wang W, Xiao L, Li R, **Luo S**. (2026). A functional joint model for survival and multivariate sparse functional data in multi-cohort Alzheimer's disease study, *Statistics in Medicine*, 45(3-5):e70442, PMID: 41684276, PMCID: PMC12902813.
61. Alam MS*, Kaddurah-Daouk R, **Luo S**[†]. (2026). Joint modeling of high-dimensional longitudinal data and survival using supervised low-rank tensor decomposition, *Biostatistics*, in press.

Clinical papers in which Sheng Luo were either (Co-)first author or corresponding author, or played a major role

62. Anderson WF, **Luo S**, et al. (2009). Human epidermal growth factor receptor-2 and estrogen receptor expression, a demonstration project using the residual tissue repository of the Surveillance, Epidemiology, and End Results (SEER) program. *Breast Cancer Research and Treatment*, 113(1), 189-96. PMID: 18256926, PMCID: PMC2676874.
63. Yi M*, Buchholz TA, Meric-Bernstam F, Bedrosian I, Hwang RF, Ross MI, Kuerer HM, **Luo S**, Gonzalez-Angulo AM, Buzdar AU, Symmans WF, Feig BW, Lucci A, Huang EH, Hunt KK. (2011). Classification of ipsilateral breast tumor recurrences after breast conservation therapy can predict patient prognosis and facilitate treatment planning. *Annals of Surgery*, 253(3), 572-9. PMID: 21209588, PMCID: PMC4331097.
64. Yi M*, Mittendorf EA, Cormier JN, Buchholz TA, Bilimoria K, Sahin AA, Hortobagyi GN, Gonzalez-Angulo A, **Luo S**, Buzdar AU, Crow JR, Kuerer HM, Hunt KK. (2011). Novel staging system for predicting disease-specific survival in breast cancer patients treated with surgery as the first intervention: time to modify the current AJCC staging system. *Journal of Clinical Oncology*, 29(35), 4654-61. PMID: 22084362, PMCID: PMC3236648.
65. Yi M*, Meric-Bernstam F, Kuerer HM, Mittendorf EA, Bedrosian I, Lucci A, Hwang RF, Crow JR, **Luo S**, Hunt KK. (2012). Evaluation of a breast cancer nomogram for predicting risk of ipsilateral breast tumor recurrences in patients with ductal carcinoma in situ after local excision, *Journal of Clinical Oncology*, 30(6), 600-7. PMID: 22253459, PMCID: PMC3295558.
66. Park M**, **Luo S**, Kwon J, Stock TH, Delclos G, Kim H, Yun-Chul H. (2013). Effects of air pollution on asthma hospitalization rates in different age groups in metropolitan cities of Korea, *Air Quality, Atmosphere and Health*. 6:543-51, PMID: 24223075, PMCID: PMC3821782.
67. Simuni T, **Luo S**, Chou KL, Fernandez H, He B*, Parashos S. (2013). Rankin scale as a potential measure of global disability in Parkinson's disease, *Journal of Clinical Neuroscience*, 20(9), 1200-3, PMID: 23810387, PMCID: PMC3947899.
68. Parashos SA, **Luo S**, Biglan KM, Bodis-Wollner I, He B*, Liang GS, Ross W, Tilley BC, Shulman LM, on behalf of the NINDS NET-PD Investigators. (2014). Measuring disease progression in early Parkinson disease: the National Institutes of Health Exploratory Trials in Parkinson Disease (NET-PD) experience, *JAMA-Neurology*, 71(6), 710-6, PMID: 24711047, PMCID: PMC4188544.
69. Goetz CG, **Luo S**, Wang L*, Tilley BC, LaPelle NR, Stebbins GT. (2015). Handling missing values in the MDS-UPDRS, *Movement Disorders*, 30(12):1632-8, PMID: 25649812, PMCID: PMC5072275.
70. Kieburtz K, the NINDS NET-PD Investigators (including **Luo S**). (2015). Effect of creatine monohydrate on clinical progression in patients With Parkinson disease: A randomized clinical trial, *JAMA*, 313(6):584-93, PMID: 25668262, PMCID: PMC4349346.
71. The NINDS NET-PD FS-ZONE Investigators (including **Luo S**). (2015). Pioglitazone in early Parkinson's disease: a phase 2, multicentre, double-blind, randomised trial, *The Lancet Neurology*, 14(8):795-803, PMID: 26116315, PMCID: PMC4574625.

72. Goetz CG, Liu Y*, Stebbins GT, Wang L*, Tilley BC, **Luo S**, (2016). Gender-, age- and race/ethnicity-based Differential Item Functioning (DIF) analysis of MDS-UPDRS, *Movement Disorders*, 31(12):1865-73, PMID: 27943473, PMCID: PMC5289884.
73. Martino D, Pringsheim TM, Cavanna AE, Colosimo C, Hartmann A, Leckman JF, **Luo S**, Munchau A, Goetz CG, Stebbins GT, Martinez-Martin P. (2017). Systematic review of severity scales and screening instruments for tics: critique and recommendations, *Movement Disorders*, 32(3):467-73, PMID: 28071825, PMCID: PMC5482361.
74. Li K*, Chan W, Doody RS, Quinn J, **Luo S**[†]. (2017). Prediction of conversion to Alzheimer's disease with longitudinal measures and time-to-event data, *Journal of Alzheimer's Disease*, 58(2):361-71, PMID: 28436391, PMCID: PMC5477671, press release link: http://www.iospress.nl/ios_news/prediction-of-conversion-to-alzheimers-disease-with-longitudinal-measures-and-time-to-event-data
75. **Luo S**[†], Liu Y*, Teresi J, Stebbins GT, Goetz CG. (2017). Differential item functioning in the Unified Dyskinesia rating scale (UDysRS), *Movement Disorders*, 32(8):1244-9, PMID: 28568607, PMCID: PMC5570621.
76. Li K*, Furr-Stimming E, Paulsen JS, **Luo S**[†]. (2017). Dynamic prediction of motor diagnosis in Huntington's disease using a joint modeling approach, *Journal of Huntington's Disease*, 6(2):127-37, PMID: 28582868, PMCID: PMC5505650.
77. Li K*, O'Brien R, Lutz M, **Luo S**[†], (2018). A prognostic model of Alzheimer's disease relying on multiple longitudinal measures and time-to-event data, *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*, 14(5):644-51, PMID: 29306668, PMCID: PMC5938096.
78. **Luo S**[†], Ren X*, Han W, Goetz CG, Stebbins GT. (2018). Missing Data in the Unified Dyskinesia Rating Scale (UDysRS), *Movement Disorders Clinical Practice*, 5(5):523-6, PMID: 30515442, PMCID: PMC6207127.
79. Li F*, Li K, Li C*, **Luo S**[†]. (2019). Predicting the risk of Huntington's disease with multiple longitudinal biomarkers, *Journal of Huntington's Disease*, 8(3):323-32, PMID: 31256145, PMCID: PMC6718328.
80. Ren X*, Lin J*, **Luo S**[†], Goetz CG, Stebbins GT, Cubo E. (2019). Successful use of the Unified Dyskinesia Rating Scale regardless of PD- or dyskinesia-duration, *Parkinsonism & Related Disorders*, 67:113-6, PMID: 31495732, PMCID: PMC6858562.
81. Li JB, **Luo S** (co-first author), Wang MCS, Li C, Feng LF, Peng J, Li JH, Zhang X, (2019). Longitudinal associations between BMI change and the risks of colorectal cancer incidence, cancer-related and all-cause mortality among 81,388 older adults, *BMC Cancer*, 19(1), 1082, PMID: 31711465, PMCID: PMC6844049.
82. Lutz MW, **Luo S**, Williamson DE, Chiba-Falek O, (2020). Shared Genetic Etiology Underlying Late-Onset Alzheimer's Disease and Post Traumatic Stress Syndrome, *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*, 16:1280-1292, PMID: 32588970, PMCID: PMC7769164.
83. Goetz CG, Stebbins GT, **Luo S**, (2020). Movement Disorder Society-Unified Parkinson's Disease Rating Scale Use in the COVID-19 Era. *Movement Disorders*, 35(6), 911, PMID: 32320498, PMCID: PMC7264595.
84. Mao R*, Li K, Cai JQ, **Luo S**, Turner M, Blazer D, Zhao H. (2021). Adjuvant chemotherapy versus observation following resection for patients with nonmetastatic poorly differentiated colorectal neuroendocrine carcinomas, *Annals of Surgery*, 274(2):e126-e133, PMID: 31478977, PMCID: PMC12969781.

85. Ren X*, Lin J*, Stebbins GT, Goetz CG, **Luo S**[†]. (2021). Prognostic modeling of Parkinson's disease progression using early longitudinal patterns of change, *Movement Disorders*, 36(12):2853-2861, PMID: 34327755, PMCID: PMC8688189.
86. Liu H, Lutz M, **Luo S**[†]. (2021). Association between polygenic risk score and the progression from mild cognitive impairment to Alzheimer's dementia, *Journal of Alzheimer's Disease*, 84:1323-1335, PMID: 34657885, PMCID: PMC8957404.
87. **Luo S**, Zou H*, Goetz CG, Choi D*, Oakes D, Simuni T, Stebbins GT. (2021). Novel approach to MDS-UPDRS monitoring in clinical trials: longitudinal item response theory models, *Movement Disorders Clinical Practice*, 8(7):1083-1091, PMID: 34631944, PMCID: PMC8485609.
88. Sikich L, Kolevzon A, King BH, McDougale CJ, Sanders KB, Kim SJ, Spanos M, Chandrasekhar T, Del Pilar Trelles M, Rockhill CM, Palumbo ML, Cundiff AW, Montgomery A, Siper P, Minjarez M, Nowinski LA, Marler S, Shuffrey LC, Alderman C, Weissman J, Zappone B, Mullett JE, Crosson H, Hong N, Siecinski SK, Giamberardino SN, **Luo S**, She L, Bhapkar M, Dean R, Scheer A, Johnson JL, Gregory SG, Veenstra-VanderWeele J, (2021). Intranasal oxytocin in children and adolescents with autism spectrum disorder, *New England Journal of Medicine*, 385(16):1462-1473, PMID: 34644471, PMCID: PMC9701092.
89. Wang SH, Guo Y*, Ervin JF, Lusk JB, **Luo S**. (2022). Neuropathological associations of limbic-predominant age-related TDP-43 encephalopathy neuropathological change (LATE-NC) differ between the oldest-old and younger-old, *Acta Neuropathologica*, 144(1):45-57, PMID: 35551470, DOI: 10.1007/s00401-022-02432-5.
90. **Luo S**, Goetz CG, Choi D*, Aggarwal S, Mestre TA, Stebbins GT. (2022). Resolving missing data from the MDS-UPDRS: implications for telemedicine, *Movement Disorders*, 37(8):1749-1755, PMID: 35716143, PMCID: PMC9391277.
91. **Luo S**, Zou H*, Stebbins GT, Schwarzschild MA, Macklin EA, Chan J, Oakes D, Simuni T, Goetz CG. (2022). Dissecting the domains of PD: insights from longitudinal Item Response Theory modeling, *Movement Disorders*, 37(9):1904-1914, PMID: 35841312, PMCID: PMC9897939.
92. Liu H, Lutz M, **Luo S**[†]. (2022). Genetic association between epigenetic aging-acceleration and the progression of mild cognitive impairment to Alzheimer's disease, *Journals of Gerontology: Biological Sciences*, 77(9):1734-1742, PMID: 35797594, PMCID: PMC9434458.
93. Zou H*, Aggarwal V, Stebbins GT, Muller M, Cedarbaum JM, Pedata A, Stephenson D, Simuni T, **Luo S**[†]. (2022). Application of longitudinal item response theory models to modeling Parkinson's disease progression, *CPT: Pharmacometrics & Systems Pharmacology*, 11(10):1382-1392, PMID: 35895005, PMCID: PMC9574723.
94. Guo Y*, Stebbins GT, Mestre TA, Goetz CG, **Luo S**[†]. (2022). Movement Disorder Society Unified Parkinson's Disease Rating Scale motor examination retains its two-domain profile in both on and off states, *Movement Disorders Clinical Practice*, 9(8):1149-51, PMID: 36339308, PMCID: PMC9631834.
95. Zhang Y, Wang M, Chen Q* (co-first author), Deng Y, Chen J, Dai Y, **Luo S**[†] (co-corresponding author), Xu J, Zhao H, Cai J. (2022). Adverse events of immune checkpoint inhibitor-based therapies for unresectable hepatocellular carcinoma in prospective clinical trials: A systematic review and meta-analysis, *Liver Cancer*, 1-18, DOI: 10.1159/000528698.
96. Li H*, Liu H, Lutz M, **Luo S**[†]. (2023). Novel genetic variants in TP37, PIK3R1, CALM1 and PLCG2 of the neurotrophin signaling pathway are associated with the progression from mild cognitive impairment to Alzheimer's disease, *Journals of Alzheimer's Disease*, 91(3):977-987, PMID: 36530083, PMCID: PMC9905310.

97. Goetz CG, Choi D*, Guo Y*, Stebbins GT, Mestre TA, **Luo S**. (2023). It is as it was: MDS-UPDRS Part 3 scores cannot be combined with other Parts to give a valid sum, *Movement Disorders*, 38(2):342-347, PMID: 36480107, PMCID: PMC9974855.
98. Cubo E, **Luo S**, Martinez-Martin P, Stebbins GT, Lin J*, Choi D*, Garcia-Bustillo A, Mir P, Santos-Garcia D, Serrano M, Rodriguez-Violante M, Singer C, and the Spanish MDS-NMS Validation Study Group. (2023). Expanded and independent Spanish validation of the MDS-Non Motor Rating Scale, *Movement Disorders Clinical Practice*, 10(4):586-595, PMID: 37071078, PMCID: PMC10105117.
99. Kaasinen V, **Luo S**, Martinez-Martin P, Goetz CG, Stebbins GT. (2023). Cross-cultural differences in patient perceptions of dyskinesia in Parkinson's disease, *Movement Disorders*, 38(4):688-692, PMID: 36670051, doi.org/10.1002/mds.29335.
100. Guo Y*, Goetz CG, Stebbins GT, Mestre TA, **Luo S**[†]. (2023). Using MDS-UPDRS Parts 2 and 3 simultaneously: Combining the patient voice with clinician ratings, *Movement Disorders*, 38(3):453-463, PMIC: 36621935, PMCID: PMC10033355.
101. Chen Q*, Li K, Rhodin KE, Masoud SJ, Lidsky ME, Cai J, Wei Q, **Luo S**[†] (co-corresponding author), Zhao H. (2023). Primary tumor resection improves survival of gastrointestinal neuroendocrine carcinoma patients with non-resected liver metastases, *Journal of Surgical Oncology*, 127(6):945-955, PMID: 36807890.
102. Chen Q*, Rhodin KE, Li K, Kanu E, Zani S, Lidsky ME, Zhao J, Wei Q, **Luo S**[†] (co-corresponding author), Zhao H. (2023). Impact of surgical approach on short- and long-term outcomes in gastroenteropancreatic neuroendocrine carcinomas, *HPB*, 25(10):1255-1267, PMID: 37414710, DOI: 10.1016/j.hpb.2023.06.008.
103. Di Luca DG, **Luo S**, Liu H, Cohn M, Davis TL, Ramirez-Zamora, A, Rafferty MR, Dahodwala N, Naito A, Neault M, Beck JC, Marras C. (2023). Racial and ethnic differences in health-related quality of life for individuals with Parkinson's disease across Parkinson's Disease Centers of Excellence, *Neurology*, 100(21):e2170-e2081, PMID: 37019661, PMCID: PMC10238163.
104. Zou H*, Goetz CG, Stebbins GT, Schrag A, Mestre TA, **Luo S**[†]. (2023). Summing MDS-UPDRS Parts 1+2 [Non-motor and Motor Experience of Daily Living]: The Patient's Voice, *Movement Disorders*, 38(7):1363-1364, PMID: 37087725, PMCID: PMC10764068.
105. Goetz CG, Zou H*, Stebbins GT, Schrag A, Mestre TA, **Luo S**[†]. (2023). Reply to: Comment on "Summing MDS-UPDRS Parts 1+2 [Non-motor and Motor Experience of Daily Living]: The Patient's Voice", *Movement Disorders*, 38(8):1564, PMID: 37565401, PMCID: PMC10652225.
106. Mestre TA, Fabbri M, **Luo S**, Stebbins GT, Goetz CG, Sampaio C. (2023). A unified framework for evidence-based diagnostic criteria programs in movement disorders, *Movement Disorders*, 38(7):1156-1162, PMID: 37156735, DOI: 10.1002/mds.29420.
107. Mestre TA, Fabbri M, **Luo S**, Stebbins GT, Goetz CG, Sampaio C. (2023). Reply to: "The Framework for Diagnostic Criteria in Movement Disorders: The Value of Methodological Tools and Combined Criteria", *Movement Disorders*, 38(9):1763-1764, PMID: 37718268, DOI: 10.1002/mds.29588.
108. Zou H*, **Luo S**[†], Liu H, Lutz M, Bennett DA, Plassman BL, Welsh-Bohmer KA. (2023). Genotypic effects of the TOMM40'523 variant and APOE on longitudinal cognitive change over 4 years: The TOMMORROW Study, *The Journal of Prevention of Alzheimer's Disease*, 4(10):886-894, PMID: 37874111, PMCID: PMC10734664.
109. Chen Q*, Rhodin KE, Li K, Kanu E, Zani S, Lidsky ME, Zhao J, Wei Q, **Luo S**[†] (co-corresponding author), Zhao H. (2024). Impact of primary tumor resection and metastasectomy among gastroenteropancreatic neuroendocrine tumors with liver metastases only on survival, *HPB*, 26(1):125-136, PMID: 37806829, DOI: 10.1016/j.hpb.2023.09.016.

110. Zou H*, Guo Y*, Goetz CG, Mestre TA, Stebbins GT, Al-Hajraf F, Lawton M, Hu M, **Luo S**[†]. (2024). Co-existent probable RBD and PD: disease progression, medication response, and clinical trial Implications, *Movement Disorders Clinical Practice*, 11(3):312-314, PMID: 38468540, PMCID: PMC10928330.
111. Li H*, Plichta JK, Li K, Jin Y, Thomas SM, Ma F, Tang L, Wei Q, He YW, Chen Q, Guo Y, Liu Y, Zhang J, **Luo S**[†]. (2024). Impact of HER2-low status for patients with early-stage breast cancer and non-pCR after neoadjuvant chemotherapy: A National Cancer Database Analysis, *Breast Cancer Research and Treatment*, 204(1):89-105, PMID: 38066250, DOI: 10.1007/s10549-023-07171-z.
112. Nakashima M, Li K, Chen Q, de Silva S, Li H, Kawakami K, Wei Q, **Luo S**[†] (co-corresponding author), Zhao H. (2023). Appropriate dose of regorafenib based on body weight of colorectal cancer patients: A retrospective cohort study, *BMC Cancer*, 23(1):1268, PMID: 38129822, PMCID: PMC10740272.
113. Zhou X*, Zou H*, Lutz MW, Arbeev K, Akushevich I, Yashin A, Welsh-Bohmer KA, **Luo S**[†]. (2024). Assessing tilavonemab efficacy in early Alzheimer's disease via longitudinal item response theory modeling, *Alzheimer's & Dementia: Transnational Research & Clinical Interventions*, 10(2):e12471, PMID: 38835820, PMCID: PMC11148533.
114. Li H*, Wu Y, Zou H*, Koner S*, Plichta JK, Tolaney SM, Zhang J, He YW, Wei Q, Tang L, Zhang H, Zhang B, Guo Y*, Chen X, Li K, Lian L, Ma F, **Luo S**[†]. (2024). Clinical efficacy of CDK4/6 inhibitor plus endocrine therapy in HR-positive/HER2-0 and HER2-low-positive metastatic breast cancer: a secondary analysis of PALOMA-2 and PALOMA-3 trials, *eBioMedicine*, 105:105186, PMID: 38861871, PMCID: PMC11215206.
115. Chen Q*, Li K, Rhodin KE, Deng Y, Lidsky ME, **Luo S**[†] (co-corresponding author), Ding P. (2024). Development and internal validation of individualized prediction models of overall survival and 6-Month mortality among patients with synchronous early-onset colorectal liver metastases, *HPB*, 26(11):1349-1363, PMID: 39122641, DOI: 10.1016/j.hpb.2024.07.413.
116. Zou H*, Stebbins GT, Simuni T, **Luo S**, Cedarbaum JM. (2024). Validating new symptom emergence as a patient-centric outcome measure for PD clinical trials, *Parkinsonism and Related Disorders*, 128, 107118, PMID: 39353265, PMCID: PMC12418693.
117. Zou H*, Goetz CG, Stebbins GT, Mestre TA, **Luo S**[†], (2025). Increasing sensitivity in patient-reported MDS-UPDRS items for predicting medication initiation in early PD, *Movement Disorders Clinical Practice*, 12(2):148-156, PMID: 39422345, PMCID: PMC11802657.
118. Zou H*, Lutz MW, Welsh-Bohmer K, **Luo S**[†], (2025). Bayesian integration of longitudinal and survival outcomes in Alzheimers prediction, *Alzheimer's & Dementia*, 21(9):e70094, PMID: 40891267, PMCID: PMC12402701.
119. Yao S*, Shang Q*, Ouyang M*, Zhou H*, Yao Z, Liu Y, **Luo S**[†], (2025). Designing single-arm clinical trials: principles, applications, and methodological considerations, *Annals of Clinical Epidemiology*, 7(3):90-98, PMID: 40697795, PMCID: PMC12279405.
120. Shang Q*, Yao S*, Ouyang M*, Wang X, **Luo S**[†], (2025). Secondary analysis of randomized controlled trials: Methodological considerations and best practices, *Annals of Clinical Epidemiology*, 7(4):137-147, PMID: 41393867, PMCID: PMC12699228.
121. Ghosh D*, Pal, S, Lutz MW, **Luo S**[†], (2025). Ensemble survival analysis for preclinical cognitive decline prediction in Alzheimer's disease using longitudinal biomarkers, *Journal of Alzheimer's Disease*, 107(3):1256-1266, PMID: 40808576, PMCID: PMC12421452.
122. Alam MS*, Yu L*, Stebbins GT, Mestre TA, Goetz CG, **Luo S**[†], (2025). Longitudinal evaluation of an abbreviated patient-reported MDS-UPDRS for predicting dopaminergic therapy initia-

- tion in early Parkinson's disease, *Movement Disorders*, 41(2):426-435, PMID: 41144709, PMCID: PMC12597095.
123. Liu B*, Shang Q*, Li J, Yao S*, Ouyang M*, Wang Y, **Luo S**[†], Ouyang Q (2026). AI and big Data in oncology: A physician-centered perspective on emerging clinical and research applications, *Cancer Innovation*, 5(1):e70047, PMID: 41624314, PMCID: PMC12855167.
 124. Ghosh D*, Tushar F, Dahal L, Vancoillie L, Lafata KJ, Samei E, Lo JY, **Luo S**[†], (2026). Demographic distribution matching between real world and virtual phantom population, *Medical Physics*, 53(3):e70364, PMID: 41746164, DOI: 10.1002/mp.70364.
 125. Zou H*, Wang T, Huynh K, Meikle P, Kaddurah-Daouk R, **Luo S**[†], (2026). Longitudinal plasma lipidomics enhance AD conversion prediction in the ADNI cohort, *Journal of Alzheimer's Disease*, in press.
 126. D'Anniballe VM, Kim S, Finlay JB, Wang M, Ko T, **Luo S**, Whitson HE, Johnson KG, Goldstein BJ (2026). Olfactory cleft biopsy analysis of Alzheimers disease pathobiology across disease stages, *Nature Communications*, 17, published online on March 18, 2026.
 127. Shang Q*, Lin Z*, Plichta J, Thomas S, Ouyang M*, **Luo S**[†], Wang X, (2026). Comprehensive analysis and prediction of HER2-targeted therapy insensitivity among HER2-positive breast cancer patients undergoing neoadjuvant treatment, *Cancers*, 18(6), 989, <https://doi.org/10.3390/cancers18060989>.

Collaborative papers

128. Elm JJ, the NINDS NET-PD Investigators (including **Luo S**). (2012). Design innovations and baseline findings in a long-term Parkinson's trial: the National Institute of Neurological Disorders and Stroke Exploratory Trials in Parkinson's Disease Long-Term Study-1. *Movement Disorders*, 27(12), 1513-21. PMID: 23079770, PMCID: PMC3481179.
129. Goetz CG, Stebbins GT, Wang L*, LaPelle NR, **Luo S**, Tilley BC. (2014). MDS-sponsored scale translation program: process, format, and clinimetric testing plan for the MDS-UPDRS and UDysRS, *Movement Disorders Clinical Practice*, 1(2), 97-101, PMID: 27747259, PMCID: PMC5065071.
130. Kashiwara K, Kondo T, Mizuno Y, Kikuchi S, Kuno S, Hasegawa K, Hattori N, Mochizuki H, Mori H, Murata M, Nomoto M, Takahashi R, Takeda A, Tsuboi Y, Ugawa Y, Yamanmoto M, Yokochi F, Yoshii F, Stebbins GT, Tilley BC, **Luo S**, Wang L*, LaPelle NR, Goetz CG; MDS-UPDRS Japanese Validation Study Group. (2014). Official Japanese version of the movement disorder society-unified Parkinson's disease rating scale: validation against the original English version, *Movement Disorders Clinical Practice*, 1(3), 200-12, PMID: 25328906, PMCID: PMC4199098.
131. Cubo E, Goetz CG, Stebbins GT, LaPelle NR, Tilley BC, Wang L*, **Luo S**, on behalf of the Spanish UDysRS Program Members. (2014). Independent Spanish validation of the unified Dyskinesia rating scale, *Movement Disorders Clinical Practice*, 1(3), 213-8, PMID: 27891530, PMCID: PMC5123681.
132. Walters AS, et al., and members of the MDS Committee on Rating Scales (including **Luo S**). (2014). Review of quality of life instruments for the restless legs syndrome/Willis-Ekbom disease (RLS/WED): Critique and recommendations, *Journal of Clinical Sleep Medicine*, 10(12), 1351-7, PMID: 25348243, PMCID: 4237531.
133. Walters AS, et al., and members of the MDS Committee on Rating Scales (including **Luo S**). (2014). Review of diagnostic instruments for the restless legs syndrome/Willis-Ekbom disease (RLS/WED): Critique and recommendations, *Journal of Clinical Sleep Medicine*, 10(12), 1343-9, PMID: 25348242, PMCID: 4237530.

134. Brown AL**, Poston WSC, Jahnke SA, Haddock CK, **Luo S**, Delclos GL, Day RS. (2015). Weight advice associated with male firefighter weight perception and behavior, *American Journal of Preventive Medicine*, 49(4):589-93, PMID: 26141913, PMCID: PMC5059839.
135. Bega D, **Luo S**, Fernandez H, Chou K, Aminoff M, Glazmann S, Huang M*, Houston E, Parashos S, Walker H, Russell DS, Christine C, Dhall R, Singer C, Bodis-Wollner I, Nicholas A, Hamill R, Truong D, Mari A, Simuni T. (2015). Impact of depression on progression of impairment and disability in early Parkinson's Disease, *Movement Disorders Clinical Practice*, 2(4):371-8, PMID: 28393083, PMCID: PMC5381520.
136. Lazaridis C, Yang M*, DeSantis SM, **Luo S**, Robertson CS. (2015). Predictors of intensive care unit length of stay and intracranial pressure in severe traumatic brain injury, *Journal of Critical Care*, 30(6):1258-62, PMID: 26324412, PMCID: PMC5040122.
137. Yu RL, Wu RM, Chan A, Mok V, Wu YR, Tilley B, **Luo S**, Wang L*, LaPelle N, Stebbins G, Goetz C. (2017). Cross-cultural differences of the non-motor symptoms studied by the traditional Chinese version of the movement disorder society-unified Parkinson's disease rating scale, *Movement Disorders Clinical Practice*, 4(1):68-77, PMID: 28345011, PMCID: PMC5365089.
138. Perez-Lloret S, et al, the Members of the MDS Committee on Rating Scales Development (including **Luo S**). (2016). Rating scales for pain in Parkinson's disease: critique and recommendations, *Movement Disorders Clinical Practice*, 3(6):527-37, PMID: 30363588, PMCID: PMC6178703.
139. Brown AL**, Poston WC, Jahnke SA, Haddock CK, **Luo S**, Delclos GP, Day RS, (2016). Weight loss advice and prospective weight change among overweight firefighters, *International Journal of Occupational and Environmental Health*, 22(3):233-9, PMID: 27467464, PMCID: PMC5102227.
140. Csencsits-Smith K, Suescn, J, Li K*, **Luo S**, Bick DL, Schiess M. (2016). Serum lymphocyte-associated cytokine concentrations change more rapidly over time in multiple system atrophy compared to Parkinson's disease, *NeuroImmunoModulation*, 23(5-6):301-8, PMID: 28395279, PMCID: PMC5491391.
141. Rafferty MR, Schmidt PN, **Luo S**, Li K*, Marras C, Davis TL, Guttman M, Cubillos F, Simuni T, on behalf of all NPF-QII Investigator. (2017). Regular exercise, quality of life, and mobility in Parkinson's disease: a longitudinal analysis of National Parkinson Foundation Quality Improvement Initiative data, *Journal of Parkinson's Disease*, 7(1):193-202, PMID: 27858719, PMCID: PMC5482526.
142. Vidoni M**, Gabriel KP, **Luo S**, Simonsick EM, Day RS. (2017). Vitamin B12 and homocysteine associations with gait speed in older adults, *Journal of Nutrition, Health, and Aging*, 21(10):1321-28, PMID: 29188896, PMCID: PMC5726303.
143. Gibbons C, Simon DK, Huang M, Tilley BC, Aminoff MJ, Bainbridge JL, Brodsky M, Freeman R, Goudreau J, Hamill RW, **Luo S**, Singer C, Videnovic A, Bodis-Wollner I, Wong PS. (2017) Autonomic and electrocardiographic findings in Parkinson's disease, *Autonomic Neuroscience: Basic and Clinical*, 205:93-8, PMID: 28506500, PMCID: PMC5609844.
144. Leehey M, **Luo S**, Sharma S, Wills AM, Bainbridge JL, Wong PS, Simon DK, Schneider J, Zhang Y*, Perez A, Dhall R, Christine CW, Singer C, Cambi F, Boyd JT. (2017). Association of metabolic syndrome and change in Unified Parkinson Disease Rating Scale scores, *Neurology*, 89(17):1789-94, PMID: 28972194, PMCID: PMC5664310.
145. Zitser J, Peretz C, Ber David A, Shabtai H, Ezra A, Kestenbaum M, Brozgol M, Rosenberg A, Herman T, Balash Y, Gadoth A, Thaler A, Stebbins GT, Goetz CG, Tilley BC, **Luo S**, Liu Y*, Giladi N, Gurevich T., (2017). Validation of the Hebrew version of the Movement Disorder Society-Unified Parkinson's Disease Rating Scale, *Parkinsonism & Related Disorders*, 45:7-12, PMID: 28965872, PMCID: PMC5972376.

146. Vidoni M**, Gabriel KP, **Luo S**, Simonsick EM, Day RS. (2018). Relationship between homocysteine and muscle strength decline: the Baltimore longitudinal study of aging, *Journal of Gerontology: Medical Sciences*, 73(4):546-51, PMID: 28958086, PMCID: PMC5861846.
147. Masha L, Stone J, Stone D, Zhang J*, **Luo S**, (2018). Pulmonary catheterization data correlate poorly with renal function in heart failure, *CardioRenal Medicine*, 8:183-191, PMID: 29635242, PMCID: PMC6170923.
148. Li K*, Yuan S, Wang W, Wan S, Ceesay P, Heyse JF, Mt-Isa S, **Luo S**. (2018). Periodic benefit-risk assessment using Bayesian stochastic multi-criteria acceptability analysis, *Contemporary Clinical Trials*, 67:100-108, PMID: 29505866, PMCID: PMC5972390.
149. Qian D, Liu H, Wang X, Ge J, **Luo S**, Patz EF, Moorman PG, Su L, Chen S, Christiani DC, Wei Q, (2019). Potentially functional genetic variants in the complement-related immunity gene-set are associated with non-small cell lung survival, *International Journal of Cancer*, 144(8):1867-1876, PMID: 30259978, PMCID: PMC6377316.
150. Chen K, Liu H, Liu Z, **Luo S**, Patz EF, Moorman PG, Su L, Shen S, Christiani DC, Wei Q, (2019). Genetic variants in RUNX3, AMD1 and MSRA in the methionine metabolic pathway and survival in non-small cell lung cancer Patients, *International Journal of Cancer*, 145(3):621-631, PMID: 30650190, PMCID: PMC6828159.
151. Yang W, Liu, H, Duan B, Xu X, Carmody D, **Luo S**, Walsh KM, Abbruzzese JL, Zhang X, Chen X, Wei Q, (2019). Three novel genetic variants in NRF2 signaling pathway genes are associated with pancreatic cancer risk, *Cancer Science*, 110(6):2022-2032, PMID: 30972876, PMCID: PMC6550126
152. Guo Y, Feng Y, Liu H, **Luo S**, Clarke JW, Moorman PG, Su L, Shen S, Christiani DC, Wei Q. (2019). Potentially functional genetic variants in the TNF/TNFR signaling pathway genes predict survival of patients with non-small cell lung cancer in the PLCO cancer screening trial. *Molecular Carcinogenesis*, 58(7):1094-1104, PMID: 30989732, PMCID: PMC6548610.
153. Xu X, Liu H, Cruz D, **Luo S**, Walsh KM, Abbruzzese JL, Zhang X, Wei Q, (2019). Genetic variants in the liver kinase B1-AMP-activated protein kinase pathway genes and pancreatic cancer risk, *Molecular Carcinogenesis*, 58(8):1338-48, PMID: 30997723, PMCID: PMC6602843.
154. Ge J, Liu H, Qian D, Wang X, Moorman PG, **Luo S**, Hwang S, Wei Q, (2019). Genetic variants of genes in the NER pathway associated with risk of breast cancer: a large-scale analysis of 14 published GWAS datasets in the DRIVE Study, *International Journal of Cancer*, 145(5), 1270-9, PMID: 31026346, PMCID: PMC6930956.
155. Mao R*, Zhao H, Li K, **Luo S**, Turner M, Cai JQ, Blazer D. (2019). Outcomes of lymph node dissection for non-metastatic pancreatic neuroendocrine tumors: a propensity score-weighted analysis of the National Cancer Database, *Annals of Surgical Oncology*, 26(9):2722-9, PMID: 31209670, PMCID: PMC7436555.
156. Dai W, Liu H, Xu X, Jie G, **Luo S**, Zhu D, Amos CI, Fang S, Lee JE, Li, X, Nan H, Li C, Wei Q, (2019). Genetic variants in ELOVL2 and HSD17B12 predict melanoma-specific survival, *International Journal of Cancer*, 145(10), 2619-28, PMID: 30734280, PMCID: PMC6824721.
157. Amrhein TJ, Bozdogan E, Vekaria S, Patel P, Lerebours R, **Luo S**, Kranz PG. (2019). Cross-sectional CT assessment of the extent of injectate spread at CT fluoroscopy-guided cervical epidural interlaminar steroid injections. *Radiology*, 292(3):723-9, PMID: 31310176, PMCID: PMC6716562.
158. Schwartz FR, Shaw BI, Lerebours R, Vernuccio F, Rigioli F, Gonzalez F, **Luo S**, Rege AS, Vikraman D, Hurwitz-Koweek L, Marin, D, Ravindra K. (2020). Correlation of preoperative imaging characteristics with donor outcomes and operative difficulty in laparoscopic donor nephrectomy. *American Journal of Transplantation*, 20(3):752-60, PMID: 31553125, PMCID: PMC7042043.

159. Wang Z, Bier EA, Swaminathan A, Parikh K, Nouns J, He M, Mammarappallil JG, **Luo S**, Driehuys B, Rajagopal S, (2019). Diverse cardiopulmonary diseases are associated with distinct Xenon MRI signatures. *European Respiratory Journal*, 54(6):1900831, PMID: 31619473, PMCID: PMC7271066 .
160. He M, Wang Z, Rankine L, **Luo S**, Nouns J, Virgincar R, Mammarappallil JG, Driehuys B, (2020). Generalized linear binning to compare hyperpolarized ¹²⁹Xe ventilation maps derived from 3D radial gas exchange versus dedicated multi-slice gradient Echo MRI. *Academic Radiology*, 27(8):e193-e203, PMID: 31786076, PMCID: PMC7250711.
161. Chaudhry M, McGinty KA, Mervak B, Lerebours R, Li C, Shropshire E, Ronald J, Commander L, Hertel J, **Luo S**, Bashir MR, Burke LMB. (2020). The LI-RADS v2018 Treatment Response algorithm is accurate for evaluating ablated hepatocellular carcinoma. *Radiology*, 294(2):320-6, PMID: 31845843, DOI: 10.1148/radiol.2019191581.
162. Tang D, Zhao Y, Liu H, **Luo S**, Clarke JW, Glass C, Su L, Shen S, Christiani DC, Gao W, Wei Q, (2020). Novel genetic variants in HDAC2 and PPARGC1A genes of the CREB-binding protein pathway predict survival of non-small cell lung cancer, *Molecular Carcinogenesis*, 59:104-115, PMID: 31713888, PMCID: PMC7481022.
163. Yang S, Tang D, Zhao Y, Liu H, **Luo S**, Stinchcombe TE, Glass C, Su L, Shen S, Christiani DC, Wang Q, Wei Q, (2020). Novel genetic variants in KIF16B and NEDD4L in the endosome-related genes are associated with non-small cell lung cancer survival, *International Journal of Cancer*, 147(2):392-403, PMID: 31618441, PMCID: PMC8096203.
164. Dai W, Liu H, Liu Y, Xu X, Qian D, **Luo S**, Cho E, Zhu D, Amos CI, Fang S, Lee JE, Li X, Nan H, Li C, Wei Q, (2020). Genetic variants in the folate metabolic pathway genes predict melanoma-specific survival. *British Journal of Dermatology*, 183(4):719-728, PMID: 31955403, PMCID: PMC7367702.
165. Zhou B, Zhao Y, Liu H, **Luo S**, Amos CI, Lee JE, Li X, Nan H, Wei Q, (2020). Novel genetic variants of ALG6 and GALNTL4 of the glycosylation pathway predict cutaneous melanoma-specific survival, *Cancers*, 12(2):288, PMID: 31991610, PMCID: PMC7072252.
166. Tang D, Zhao Y, Liu H, **Luo S**, Clarke JM, Glass C, Su L, Shen S, Christiani DC, Gao W, Wei Q, (2020). Potentially functional genetic variants in PLIN2, SULT2A1 and UGT1A9 genes of the Ketone pathway and survival of non-small cell lung cancer, *International Journal of Cancer*, 147(6):1559-1570, PMID: 32072637, PMCID: PMC8078192.
167. Liu X, Qian D, Liu H, Abbruzzese JL, **Luo S**, Walsh KM, Wei Q, (2020). Genetic variants of the peroxisome proliferator-activated receptor (PPAR) signaling pathway genes and risk of pancreatic cancer, *Molecular Carcinogenesis*, 59(8):930-939, PMID: 32367578, PMCID: PMC7592725.
168. Deng W, Liu H, **Luo S**, Clarke J, Glass C, Su L, Lin L, Christiani DC, Wei Q, (2020). APOB genotypes and CDH13 haplotypes in the cholesterol-related pathway genes predict non-small cell lung cancer survival, *Cancer Epidemiology, Biomarkers & Prevention*, 29(6):1204-1213, PMID: 32238407, PMCID: PMC7269811.
169. Dai W, Liu H, Chen K, Xu X, Qian D, **Luo S**, Amos CI, Lee JE, Li X, Nan H, Li C, Wei Q, (2020). Genetic variants in PDSS1 and SLC16A6 in the ketone body metabolic pathway predict cutaneous melanoma-specific survival, *Molecular Carcinogenesis*, 59(6), 640-650, PMID: 32232919, PMCID: PMC7454142.
170. Faust-Socher A, Anis S, Kestenbaum M, Shabtai H, Taichman T, David AB, Ezra A, Peretz C, Rosenberg A, Brozgov M, Herman T, Stebbins GT, Goetz CG, Martinez-Martin P, **Luo S**, Ren X*, Hausdorff J, Giladi N, Gurevich T, (2020). Validation of the Hebrew version of the Unified Dyskinesia Rating Scale (UDysRS), *Neuroepidemiology*, 54(4):356-362, PMID: 32541146, DOI: 10.1159/000507827.

171. Zhao Y, Tang D, Yang S, Liu H, **Luo S**, Stinchcombe TE, Glass C, Su L, Shen S, Christiani DC, Wei Q, (2020). Novel variants of ELP2 and PIAS1 in the interferon gamma signaling pathway are associated with non-small cell lung cancer survival, *Cancer Epidemiology, Biomarkers & Prevention*, 29(8):1679-1688, PMID: 32493705, PMCID: PMC7415606.
172. Tosin MH, Goetz CG, Choi D*, **Luo S**, Stebbins GT, (2020). Item Response Theory Analysis of the MDS-UPDRS Motor Examination: Tremor vs. Non-tremor Items. *Movement Disorders*, 35(9):1587-95, PMID: 32469456, DOI:10.1002/mds.28110.
173. George SZ, Li C*, **Luo S**, Horn ME, Lentz TA, (2020). Longitudinal monitoring of pain associated distress with the optimal screening for prediction of referral and outcome yellow flag (OSPRO-YF) tool: predicting reduction pain intensity and disability, *Archives of Physical Medicine and Rehabilitation*, 101(10):1763-1770, PMID: 32599059, PMCID: PMC7078059.
174. Gu N, Dai W, Liu H, Ge J, **Luo S**, Cho E, Amos CI, Lee JE, Li X, Nan H, Yuan H, Wei Q, (2020). Genetic variants in TKT and DERA in the NADPH Pathway predict melanoma survival, *European Journal of Cancer*, 136:84-94, PMID: 32659474, PMCID: PMC7540967.
175. Wu Y, Yang S, Liu H, **Luo S**, Stinchcombe TE, Glass C, Su L, Shen S, Christiani DC, Wang Q, Wei Q, (2020). Novel genetic variants of KIR3DL2 and PVR involved in immunoregulatory interactions are associated with non-small cell lung cancer survival, *American Journal of Cancer Research*, 10(6):1770-84, PMID: 32642289, PMCID: PMC7339263.
176. Zhao L, Liu H, **Luo S**, Walsh KM, Li W, Wei Q, (2020). Associations of novel variants in PIK3C3, INSR and MAP3K4 of the ATM pathway genes with pancreatic cancer risk, *American Journal of Cancer Research*, 10(7):2128-2144, PMID: 32775006, PMCID: PMC7407350.
177. Siuda J, Boczarska-Jedynak M, Budrewicz S, Figura M, Fiszer U, Gajos A, Gorzkowska A, Kozirowska-Gawron E, Kozirowski D, Krygowska-Wajs A, Rudzinska-Bar M, Slawek J, Ren X*, Luo S, Martinez-Martin P, Stebbins G, Goetz CG, Opala G, Bogucki A, Dulski J, Janik P, Koszewicz M, Lenska-Mieciek M, Michalowska M, Piascik-Gromada M, Potasz-Kulikowska K, Smilowski M, Wasielewska A, Wjciak-Pedziwiatr M, (2020). Validation of the Polish version of the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS), *Polish Journal of Neurology and Neurosurgery*, 54(5):416-425, PMID: 32639019, doi: 10.5603/PJNNS.a2020.0049.
178. Qian D, Liu H, Zhao L, Wang X, **Luo S**, Moorman PG, Patz EF, Su L, Shen S, Christiani DC, Wei Q, (2020). Novel genetic variants in genes of the Fc gamma receptor-mediated phagocytosis pathway predict non-small cell lung cancer survival, *Translation Lung Cancer Research*, 9(3):575-586, PMID: 32676321, PMCID: PMC7354140.
179. Wang H, Zhao L, Liu H, **Luo S**, Akinyemiju T, Hwang S, Wei Q, (2020). Variants in SNAI1, AMDHD1 and CUBN in vitamin D pathway genes are associated with breast cancer risk: a large-scale analysis of 14 GWASs in the DRIVE study, *American Journal of Cancer Research*, 10(7):2160-2173, PMID: 32775008, PMCID: PMC7407344.
180. Liu L, Liu H, **Luo S**, Patz EF, Glass C, Su L, Lin L, Christiani DC, Wei Q, (2020). Novel genetic variants of SYK and ITGA1 related lymphangiogenesis signaling pathway predict non-small cell lung cancer survival, *American Journal of Cancer Research*, 10(8):2603-2616, PMID: 32905494, PMCID: PMC7471352.
181. Tang D, Liu H, Zhao Y, Qian D, **Luo S**, Patz EF, Su L, Shen S, Christiani DC, Gao W, Wei Q, (2020). Genetic variants of BIRC3 and NRG1 in the NLRP3 inflammasome pathway are associated with non-small cell lung cancer survival, *American Journal of Cancer Research*, 10(8):2582-2595, PMID: 32905523, PMCID: PMC7471354.
182. Zhao L, Liu H, **Luo S**, Moorman PG, Walsh KM, Li W, Wei Q, (2020). Associations between genetic variants of KIF5B, FMN1, and MGAT3 in the cadherin pathway and pancreatic cancer risk, *Cancer Medicine*, 9(24):9620-9631, PMID: 33200553, PMCID: PMC7774717.

183. Lu G, Zhou B, Zhao YC, Liu H, **Luo S**, Amos CI, Lee JE, Yang K, Qureshi A, Han J, Wei Q, (2020). Novel genetic variants of PIP5K1C and MVB12B of the endosome-related pathway predict cutaneous melanoma-specific survival, *American Journal of Cancer Research*, 10(10):3382-3394, PMID:33163277, PMCID: PMC7642651.
184. Wang H, Liu H, Dai W, **Luo S**, Amos CI, Lee JE, Li X, Yue Y, Nan H, Wei Q, (2020). Association of genetic variants of TMEM135 and PEX5 in the peroxisome pathway with cutaneous melanoma-specific survival, *Annals of Translational Medicine*, 9(5):396, PMID: 33842617, PMCID: PMC8033299.
185. Wu Y, Liu Z, Tang D, Liu H, **Luo S**, Stinchcombe TE, Glass C, Su L, Lin L, Christiani DC, Wang Q, Wei Q, (2020). Potentially functional variants of HBEGF and ITPR3 in GnRH signaling pathway genes predict survival of non-small cell lung cancer, *Translational Research*, 233:92-103, PMID: 33400994, PMCID: PMC8184605.
186. Wildman-Tobriner B, Ngo L, Jaffe TA, Ehieli WL, Ho LM, Lerebours R, **Luo S**, Allen BC, (2021). Automated structured reporting for thyroid ultrasound: effect on reporting errors and efficiency, *Journal of the American College of Radiology*, 18(2):265-273, PMID: 32818484.
187. Siuda J, Boczerska-Jedynak M, Budrewicz S, Dulski, J, Figura M, Fiszer U, Gajos A, Gorzkowska A, Koziorowska-Gawron E, Koziorowski D, Krygowska-Wajs A, Rudzinska-Bar M, Slawek J, Tos M, Wojcik-Pedziwiatr M, Lin J*, Luo S, Martinez-Martin P, Stebbins GT, Goetz CG, Bogucki A, Janik P, Koszewicz M, Lenska-Mieciek M, Michalowska M, Piascik-Gromada M, Potasz-Kulikowska K, Smilowski M, Wasielewska A, Opala G, (2021). Validation of the Polish version of the Unified Dyskinesia Rating Scale (UDysRS), *Polish Journal of Neurology and Neurosurgery*, 55(2):186-194, PMID: 33528833, DOI: 10.5603/PJNNS.a2021.0010.
188. Yang S, Tang D, Zhao YC, Liu H, **Luo S**, Stinchcombe TE, Glass C, Su L, Shen S, Christiani DC, Wang Q, Wei Q, (2021). Potentially functional variants of ERAP1, PSMF1 and NCF2 in the MHC-I-related pathway predict non-small cell lung cancer survival, *Cancer Immunology, Immunotherapy*, 70(10):2819-2833, PMID: 33651148, PMCID: PMC8410895.
189. Du H, Liu L, Liu H, **Luo S**, Patz EF, Glass C, Su L, Lin L, Christiani DC, Wei Q, (2021). Genetic variants of DOCK2, EPHB1 and VAV2 in the natural killer cell-related pathway are associated with non-small cell lung cancer survival, *American Journal of Cancer Research*, 11(5):2264-2277, PMID: 34094683, PMCID: PMC8167686.
190. Mu R, Liu L, Liu H, **Luo S**, Patz EF, Glass C, Su L, Du M, Christiani DC, Jin L, Wei Q, (2021). Genetic variants of CHEK1, PRIM2 and CDK6 in the mitotic phase-related pathway are associated with non-small cell lung cancer survival, *International Journal of Cancer*, 149(6):1302-12, PMID: 34058013, PMCID: PMC8916708.
191. Horn ME, George SZ, Li C*, **Luo S**, Lentz TA. (2021). Derivation of a risk assessment tool for prediction of long-term pain intensity reduction after physical therapy, *Journal of Pain Research*, 14:1515-1524, PMID: 34093037, PMCID: PMC8169054.
192. He Y, Liu H, **Luo S**, Amos CI, Lee JE, Yang K, Han J, Wei Q, (2021). Genetic variants of EML1 and HIST1H4E in myeloid cell-related pathway genes independently predict cutaneous melanoma-specific survival, *American Journal of Cancer Research*, 11(6):3252-3262, PMID: 34249459, PMCID: PMC8263692.
193. George SZ, Yan X*, **Luo S**, Olson SA, Reinke EK, Bolognesei MP, Horn ME. (2021). Can PROMIS measures be used to create subgroups for patients seeking orthopaedic care, *Bone & Joint Open*, 2(7):493-502, PMID: 34251274, PMCID: PMC8325967.
194. Wang H, Liu L, Zhao L, **Luo S**, Akinyemiju T, Hwang S, Yue Y, Wei Q, (2021). Association of genetic variants of FBXO32 and FOXO6 in FOXO pathway with breast cancer risk, *Molecular Carcinogenesis*, 60(10):661-670, PMID: 34197655, PMCID: PMC8475729.

195. Kaasinen V, Scheperjans F, Krpp M, Korpela J, Brck A, Sipil J, Joutsa J, Jrvel J, Eerola-Rautio J, Martikainen MH, Airaksinen K, Stebbins G, Martinez-Martin P, Goetz CG, Lin J*, **Luo S**, Pekkonen E, (2021). Validation of the Finnish version of the unified dyskinesia rating scale, *European Neurology*, 84(6):444-449, PMID: 34261060, DOI: 10.1159/000517369.
196. Rigioli F, Hoyer J, Lerebours R, Lafata KJ, Li C*, Meyer M, Lyu P, Ding Y, Schwartz FR, Mettun NB, Zani S, **Luo S**, Morgan DE, Samei E, Marin D, (2021). CT radiomic features of superior mesenteric artery involvement in pancreatic ductal adenocarcinoma: a pilot study, *Radiology*, 301(3):610-622, PMID: 34491129, PMCID: PMC9899097.
197. Horn ME, Reinke EK, Yan X*, **Luo S**, Bolognisi MP, Reeve B, George SZ. (2021). Use of Patient-Reported Outcome Measurement Information System (PROMIS) measures to characterize health status for patients seeking care from an orthopedic provider: A retrospective cohort study, *BMJ Open*, 11:e047156, PMID: 34475157, PMCID: PMC8413970.
198. He Y, Liu H, **Luo S**, Amos CI, Lee, JE, Li X, Nan H, Wei Q, (2021). Genetic variants of SDCCAG8 and MAGI2 in mitosis-related pathway genes are independent predictors of cutaneous melanoma-specific survival, *Cancer Science*, 112(10):4355-4364, PMID: 34375487, PMCID: PMC8486203.
199. Liu L, Liu H, **Luo S**, Patz EF, Glass C, Su L, Lin L, Christiani DC, Wei Q, (2021). Genetic variants of CLEC4E and BIRC3 in damage-associated molecular patterns-related pathway genes predict non-small cell lung cancer survival, *Frontiers in Oncology*, 2021:3965, PMID: 34692492, PMCID: PMC8527850.
200. Hendrickson WK, Xie G*, Rahn DD, Bradley M, Sung VW, Hokanson JA, Smith AL, Visco AG, Amundsen CL, **Luo S**, Jelovsek JE. (2022). Predicting outcomes after intradetrusor onabotulinumtoxin for non-neurogenic urgency incontinence in women, *Neurology and Urodynamics*, 41:432-447, PMID: 34859485, PMCID: PMC9014828.
201. Chen AS, Liu H, Wu Y, **Luo S**, Patz EF, Glass C, Su L, Du M, Christiani DC, Wei Q, (2022). Genetic variants in DDO and PEX5L in peroxisome-related pathways predict non-small cell lung cancer survival, *Molecular Carcinogenesis*, 61(7):619-28, PMID: 35502931, DOI: 10.1002/mc.23400.
202. Jagota P, Srivannithapoom P, Petchrutchatachart S, Singmaneesakulchai S, Pisarnpong A, Lolekha P, Setthawatcharawanich S, Chairangsaris P, Limotai N, Mekawichai P, Panyakaew P, Phokaewvarangkul O, Sringean J, Pitakpatapee Y, LaPelle N, Martinez-Martin P, Ren X*, **Luo S**, Stebbins GT, Goetz CG, Bhidayasiri R, (2022). Validation of the Thai version of the MDS-UPDRS, *Journal of Movement Disorders*, 15(2):151-155, PMID: 35287261, PMCID: PMC9171300.
203. Du H, Mu R, Liu H, Wu Y, **Luo S**, Patz EF, Glass C, Su L, Du M, Christiani DC, Wei Q, (2022). Single nucleotide polymorphisms in FOXP1 and RORA of the lymphocyte activation-related pathway affect survival in lung cancer patients, *Translational Lung Cancer Research*, 11(5):890-901, PMID: 35693292, PMCID: PMC9186164.
204. Qian D, Liu H, Zhao L, **Luo S**, Walsh KM, Huang J, Li CY, Wei Q, (2022). A pleiotropic ATM variant (rs1800057 C > G) is associated with risk of multiple cancers, *Carcinogenesis*, 43:60-66, PMID: 34643693, PMCID: PMC8832460.
205. Khalil H, Aldaajani ZF, Aldughmi M, Al-Sharman A, Mohammad T, Mehanna R, El-Jaafary SI, Dahshan A, Djebara MB, Kamel WA, Amer HA, Farghal M, Abdulla F, Al-Talai N, Snineh MA, Farhat N, Jamali FA, Matar RK, Abdelraheem HS, Ghonimi NAM, Al-Melh MA, Elbhrawy S, Alotaibi MS, Elaidy SA, Almuammar SA, Al-Hashel JY, Gouider R, Samir H, Mhiri C, Skorvanek M, Lin J*, Martinez-Martin P, Stebbins G, **Luo S**, Goetz CG, Bajwa JA, (2022). Validation of the Arabic Version of the Movement Disorder Society-Unified Parkinson's Disease Rating Scale, *Movement Disorders*, 37(4):826-841, PMID: 35218056, DOI: 10.1002/mds.28905.

206. Upadhyaya S, Liu H, **Luo S**, Lutz MW, Chiba-Falek O. (2022). Polygenic risk score effectively predicts depression onset in Alzheimer's disease based on Major Depressive Disorder risk variants, *Frontiers in Neuroscience*, Vol. 16, PMID: 35350557, PMCID: PMC8957806.
207. Kannarkat GT, Rafferty MR, **Luo S**, Liu H, Mills KA. (2022). Effect of exercise and rehabilitation therapy on risk of hospitalization in Parkinson's disease, *Movement Disorders Clinical Practice*, 9(4):494-500, PMID: 35586527, PMCID: PMC9092731.
208. Lu L, Zhan M, Li X, Zhang H, Dauphars DJ, Jiang J, Yin H, Li S, **Luo S**, He Y. (2022). Clinically approved combination immunotherapy: current status, limitations, and future perspective, *Current Research in Immunology*, 3:118-127, PMID: 35676925, PMCID: PMC9167882.
209. Carey ET, Geller EJ, Rapkin A, Farb D, Cutting H, Akaninwor J, Stirling C, Bortsov A, McNulty S, Merrill P, Zakrofsky P, DeLaRosa J, Luo S, and Nackley AG, (2022). Rationale and design of a multicenter randomized clinical trial of Vestibulodynia: understanding pathophysiology and determining appropriate treatments (Vestibulodynia: UPDATE), *Annals of Medicine*, 54(1):2885-2897, PMID: 36269028, PMCID: PMC9624211.
210. Nedrud MA, Chaudhry M, Middleton MS, Moylan CA, Lerebours R, **Luo S**, Farjat A, Guy CD, Loomba R, Abdelmalek MF, Sirlin CB, Bashir M, (2023). MRI quantification of placebo effect in non-alcoholic steatohepatitis (NASH) clinical trials, *Radiology*, 306(3):e220743, PMID: 36318027, PMCID: PMC9968769.
211. French R, Hirsch J, Hemingway J, Hughes D, Lerebour R, **Luo S**, Allen B, Duszak R, Rosman. (2023). Association and trends in medicare denials and utilization for brain CT: Indirect impacts by targeted policy intervention? *Current Problems in Diagnostic Radiology*, 52(1):31-34, PMID: 35999142, DOI: 10.1067/j.cpradiol.2022.07.009.
212. Blocker SJ, Morrison S, Everitt JI, Cook J, **Luo S**, Watts T, Mowery YM, (2023). Whole-slide cytometric feature mapping for distinguishing tumor genomic subtypes in HNSCC whole slide images, *The American Journal of Pathology*, 193(2):182-190, PMID: 36414086, PMCID: PMC9885294.
213. Siecinski SK, Giamberardino SN, Spanos M, Hauser AC, Gibson JR, Chandrasekhar T, Trelles P, Rockhill CM, Palumbo ML, Cundiff AW, Montgomery A, Siper P, Minjarez M, Nowinski LA, Marler S, Kwee LC, Shuffrey LC, Alderman C, Weissman J, Zappone B, Mullett JE, Crosson H, Hong N, **Luo S**, She L, Bhapkar M, Dean R, Sheer A, Johnson JL, King BH, McDougale CJ, Sanders KB, Kim SJ, Kolevzon A, Veenstra-Vanderweele J, Hauser ER, Sikich L, Gregory SG, (2023). Genetic and epigenetic signatures associated with plasma oxytocin levels in children and adolescents with autism spectrum disorder, *Autism Research*, 16(3):502-523, PMID: 36609850, PMCID: PMC10023458.
214. Chevalier G, Udovin L, Otero-Losada M, Bordet S, Capani F, **Luo S**, Goetz CG, Perez-Lloret S, (2023). Genetics of neurogenic orthostatic hypotension in Parkinson's disease, *Brain Sciences*, 13(3):506, PMID: 36979316, PMCID: PMC10046202.
215. Bayram E, Liu H, **Luo S**, Di Luca D, Skipworth M, Solomon LD, Dahodwala N, Litvan I. (2024). Ethnoracial differences for caregiving burden in Parkinson's disease, *Parkinsonism & Related Disorders*, 118:105927, PMID: 37952272, PMCID: PMC10911683.
216. Abdraimova S, Myrzayev Z, Karimova A, Talgatkyzy A, Khaibullin T, Kaishibaeva G, Elubaeva S, Esembekova K, Choi D, Martinez-Martin P, Goetz CG, Stebbins GT, **Luo S**, Shashkin C, Zharkinbekova N, Kaiyrzhanov R (2024). Validation of the Kazakh Version of the Movement Disorder Society-Unified Parkinson's Disease Rating Scale, *Clinical Parkinsonism & Related Disorders*, 10:100232, PMID: 38292815, PMCID: PMC10826294.

217. Sotoudeh-Paima S, Segars WP, Ghosh DJ, **Luo S**, Samei E, Abadi E, (2024). A systematic assessment and optimization of photon-counting CT for lung density quantifications, *Medical Physics*, 51(4):2893-2904, PMID: 38368605, PMCID: PMC11055522.
218. Zhang K, Lin J, Li Y, Sun Y, Tong W, Li F, Chien LC, Su W, Tian H, Fu P, Qiao F, Romeiko X, Lin S, **Luo S**, Craft E, (2024). Unmasking the sky: high-resolution PM2.5 prediction in Texas using machine learning techniques, *Journal Of Exposure Science And Environmental Epidemiology*, 34(5):814-820, PMID: 38561475, DOI: 10.1038/s41370-024-00659-w.
219. Yang Q, Jiang M, Li C, **Luo S**, Crowley JM, Shaw RJ, (2024). Predicting health outcomes with intensive longitudinal data collected by mobile health devices: A functional principal component regression approach. *BMC Medical Research Methodology*, 24(1):69, PMID: 38494505, PMCID: PMC10944610.
220. Ghilardi MF, Quartarone A, DiRocco A, Calabro RS, **Luo S**, Liu H, Norcini M, Canesi M, Cian V, Zarucchi M, Ortelli P, Volpe D, Bakdounes L, Castelli D, DiFonzo A, Franco G, Frattini E, Avanzino L, Pelosin E, Ogliastro C, Ceravolo R, Palermo G, Tommasini L, Frosini D, Parnetti L, Tambasco N, Nigro P, Simoni S, Schmidt P. (2024). Supplementing best care with specialized rehabilitation treatment in Parkinson disease: A retrospective study by different expert centers, *Journal of Clinical Medicine*, 13(10):2999, PMID: 38792540, PMCID: PMC11122594.
221. Marras C, Meyer Z, Liu H, **Luo S**, Mantri S, Allen A, Baybayan S, Beck J, Brown AE, Cheung F, Dahodwala N, Davis TL, Engeland M, Fearon C, Jones N, Mills K, Miyasaki JM, Naito A, Neault M, Nelson EC, Onyinanya E, Ropa C, Weintraub D, (2024). Improving Parkinson's disease care through systematic screening for depression, *Movement Disorders Clinical Practice*, 11(10):1212-1222, PMID: 39030949, PMCID: PMC11489616.
222. Wang H, Liu H, Lu G, Tang X, **Luo S**, Du M, Christiani DC, Wei Q, (2024). Potentially functional variants of ERRFI1 in hypoxia-related genes predict survival of non-small cell lung cancer patients, *Cancer Medicine*, 13(15):e70073, PMID: 39096122, PMCID: PMC11297539.
223. Lu G, Liu H, Wang H, Tang X, **Luo S**, Du M, Christiani DC, Wei Q, (2024). Potentially functional variants of INPP5D and EXOSC3 in immunity B cell-related genes are associated with non-small cell lung cancer survival, *Frontiers in Immunology*, 15, 1440454, PMID: 39176091, PMCID: PMC11338758.
224. Lu G, Liu H, Wang H, Tang X, **Luo S**, Du M, Christiani DC, Wei Q, (2024). Genetic variants of LRRC8C, OAS2, and CCL25 in the T cell exhaustion-related genes are associated with non-small cell lung cancer survival, *Frontiers in Immunology*, 15, 1455927, PMID: 39416786, PMCID: PMC11479925.
225. Wang H, Liu H, Tang X, Lu G, **Luo S**, Du M, Christiani DC, Wei Q, (2025). Potentially functional variants of PARK7 and DDR2 in ferroptosis related genes predict survival of non-small cell lung cancer patients, *International Journal of Cancer*, 156(4):744-755, PMID: 39319523, PMCID: PMC11661923.
226. Soares T, Vale TC, Guedes LC, Maciel R, Antunes AP, Camargos S, Valadas A, Coelho M, Godinho C, Maia D, Lobo P, Maia R, Teodoro T, Rieder C, Velon AG, Rosas MJ, Calado A, Cania V, Vale J, Mendes A, Rodrigues AM, Costa MM, Lampreia T, Ferraz HB, Tumas V, Barbosa E, Stebbins GT, Tilley BC, **Luo S**, LaPelle NC, Goetz CG, Cardoso F, Ferreira JJ, (2025). Validation of the Portuguese MDS-UPDRS: challenges to obtain a scale applicable to different linguistic cultures, *Movement Disorders Clinical Practice*, 12(1):34-42, PMID: 39460979, PMCID: PMC11736879.
227. Lu G, Liu H, Wang H, **Luo S**, Du M, Christiani DC, Wei Q, (2025). Genetic variants of FER and SULF1 in the fibroblast-related genes are associated with non-small-cell lung cancer survival, *International Journal of Cancer*, 156(11):2107-2117, PMID: 39707607, PMCID: PMC11971011.

228. Vale TC, Silva DP, de Paula Brando PR, Pereira DA, Boua-Machado R, Oliveira L, de Mello Rieder CR, Barbosa ER, Cury RG, Parmera JB, Ghizoni Teive HA, Ferreira MG, Ferraz HB, da Silva CC, Tumas V, Nunes Cunha AL, Braga-Neto P, Lima DP, Della Colleta MV, Maia D, de Souza Fontana M, Almeida DO, Damas CN, Rebordo L, Lobo PP, Peralta AR, Simes R, Lampreia T, Velon AG, Raimundo R, Gago M, Mendes A, Ch NV, Damsio J, Costa S, Lopes R, Morgadinho A, Carneiro D, Magalhes A, Mestre T, Stebbins GT, **Luo S**, Goetz CG, Ferreira JJ, Cardoso F, (2025). Validation of the Portuguese version of the Movement Disorder Society Non-Motor Rating Scale (MDS-NMS) in Parkinson's disease, *Parkinsonism and Related Disorders*, 132:107276, PMID: 39848139, DOI: 10.1016/j.parkreldis.2025.107276.
229. Tushar FI, Vancoillie L, McCabe C, Kavuri A, Dahal L, Harrawood B, Fryling M, Zarei M, Sotoudeh-Paima S, Ho FC, Ghosh D, Harowicz MR, Tailor TD, **Luo S**, Segars WP, Abadi E, Lafata KJ, Lo JY, Samei E, (2025). Virtual Lung Screening Trial (VLST): An in silico study inspired by the National Lung Screening Trial for lung cancer detection. *Medical Image Analysis*, 103:103576. PMID: 40209556, PMCID: PMC12147717.
230. Dahal L, Ghoghnejad M, Vancoillie L, Ghosh D, Bhandari Y, Kim D, Ho FC, Tushar FI, **Luo S**, Lafata KJ, Abadi E, (2025). XCAT 3.0: A Comprehensive Library of Personalized Digital Twins Derived from CT scans. *Medical Image Analysis*. 2025 May 3:103636. PMID: 40347915, PMCID: PMC12247289.
231. Dahodwala N, Kapogiannis T, Cruz A, Beck J, Davis TL, Liu H, **Luo S**, Naito A, Neault M, Rafferty MR, Ramirez-Zamora A, Marras C, (2025). Longitudinal study of treatment variability for Parkinson's disease across specialized centers, *Movement Disorders Clinical Practice*, 12(12):2207-2216, PMID: 40662519, PMCID: PMC12715345.
232. Mestre TA, Stebbins GT, **Luo S**, Goetz CG, (2025). Tremor is commonly responsive to levodopa along the natural history of Parkinson disease, *Movement Disorders Clinical Practice*, 12(10):1684-1685, PMID: 40579940, DOI: 10.1002/mdc3.70162.
233. Desjardins C, Grimaldi S, **Luo S**, Yu L*, Goetz CG, Stebbins GT, Martinez-Martin P, Kurtis MM, Mestre TA, Sanchez-Ferro A, Tosin MHS, Balestrino R, Lin CY, Gasca-Salas C, Witjas T, Colin O, Maltete D, Defebvre L, Giordana C, Charif M, Thiriez C, Laurencin C, Tir M, Dupont G, Remy P, Tranchant C, Drapier S, Samier A, Benatru I, Sambin S, Corvol JC, Kelifi F, Fabbri M, Rascol O, (2026). Validation of the French translation of the Movement Disorder Society Non-Motor Symptoms Scale (MDS-NMS) in Parkinsons disease, *Movement Disorders Clinical Practice*, 13(2):575-579, PMID: 40888151, PMCID: PMC12911461.
234. Siuda J, To M, Chmiela T, Bogucki A, Borodziuk B, Budrewicz S, Dbrowski M, Gacan M, Gajos A, Gawryluk J, Janik P, Kaczyska J, Koziorowska-Gawron E, Malkiewicz J, Papu E, Kaczmarska P, Rudziska-Bar M, Sawek J, Wjcik-Pdziwiatr M, Yu L*, **Luo S**, Goetz CG, Stebbins GT, Martinez-Martin P, Mestre T, Sanchez-Ferro A, Kurtis M, Tosin MHS, Balestrino R, Lin CY, Gasca-Salas C, (2025). Validation of the Polish version of the Movement Disorder Society Non-Motor Symptoms Scale (MDS-NMS), *Neurologia i Neurochirurgia Polska, Polish Journal of Neurology and Neurosurgery*, 59(4):396-402, PMID: 40767863, DOI: 10.5603/pjnns.105278.
235. Bendig J, Frank A, Lipska-Dieck A, Wunderlich K, Geißler-Lösch D, Wurster I, Kemmner R, Brockmann K, **Luo S**, Goetz CG, Stebbins GT, Martínez-Martín P, Mestre TA, Sánchez-Ferro A, Kurtis MM, Tosin MHS, Balestrino R, Lin CY, Gasca-Salas C, Reichmann H, Falkenburger B, (2025). Validation of the German Version of the Movement Disorder Society Non-Motor Scale (MDS-NMS), *Movement Disorders Clinical Practice*, PMID: 41424371, DOI: 10.1002/mdc3.70486, published on Dec 22, 2025.
236. Alves G, Sørenes YS, Frantzen VH, Gjerstad MD, Bjørnstad AL, Maple-Grødem J, Forsaa EB, Hiorth YH, Herlofson K, Benjaminsen E, Bjørnarå KA, Dietrichs E, Balestrino R, Gasca-Salas C,

- Lin CY, Sánchez-Ferro A, Tosin MHS, Mestre TA, Kurtis MM, Martínez-Martín P, **Luo S**, Yu L*, Stebbins GT, Goetz CG, (2025). Validation of the Norwegian Version of the Movement Disorder Society–Unified Parkinson's Disease Rating Scale (MDS-UPDRS), *Clinical Parkinsonism & Related Disorders*, 14:100420, PMID: 41561570, PMCID: PMC12814832.
237. Ntowe KW, Thomas SM, Rooney MM, Dillon JL, Akinyemiju T, **Luo S**, Hwang ES, Plichta JK, (2026). Survival disparities in patients with metastatic breast cancer, *JCO Oncology Practice*, PMID: 41564368, published online on Jan 26, 2026.

Letters

1. Chen Q, Zheng QW, **Luo, S.**, Zhao H. (2021). Comment on “Resection of the primary gastrointestinal neuroendocrine tumor improves survival with or without liver treatment”. *Annals of Surgery*, 274(6):e758-e759, PMID: 33065634, doi: 10.1097/SLA.0000000000004262.
2. Chen Y, **Luo, S.** (2011). A few remarks on “Statistical distribution of the difference of two proportions” by Nadarajah and Kotz. *Statistics in Medicine*, 30(15), 1913-1915. PMID: 21671250
3. Yi M., Meric-Bernstam F., Kuerer HM, Mittendorf EA, Bedrosian I, Lucci A, Hwang RF, Crow JR, **Luo S**, Hunt, K. (2012). Reply to K Van Zee, et al for comments on “Evaluation of a breast cancer nomogram for predicting risk of ipsilateral breast tumor recurrences in patients with ductal carcinoma in Situ after local excision”, *Journal of Clinical Oncology*, 30(25), 3144-5. doi: 10.1200/JCO.2012.43.9406.

Complete List of Published Work in MyBibliography:

<http://www.ncbi.nlm.nih.gov/sites/myncbi/1De4iwaVN475n/bibliography/44658186/public/?sort=date&direction=descending>

Professional memberships

American Statistical Association (ASA)

International Chinese Statistical Association (ICSA)

The International Parkinson and Movement Disorder Society (MDS)

Institute of Mathematical Statistics (IMS)

American Association for the Advancement of Science (AAAS)

Mentoring and Education

Short Course

1. Design Clinical Trials with Non-inferiority Hypothesis: Methods and Applications, Society for Clinical Trials Annual Meeting, May 2022.
2. Design Clinical Trials with Multiple Endpoints: Methods and Applications, Society for Clinical Trials Annual Meeting, May 2021.

Faculty Mentoring Committee

1. Dr. Zhiguo Li, Associate Professor of Biostatistics and Bioinformatics, Duke University.
2. Dr. Roland Matsouaka, Assistant Professor of Biostatistics and Bioinformatics, Duke University.

3. Dr. Hwanhee Hong, Assistant Professor of Biostatistics and Bioinformatics, Duke University.

Primary Postdoctoral Fellow Mentor

1. Dr. Samsul Alam, PhD in Statistics from North Carolina State University, Jun 2024 - present.
2. Dr. Haotian Zhou, PhD in Biostatistics from University of North Carolina at Chapel Hill, Jun 2023 - present.
3. Dr. Dhrubajyoti Ghosh, PhD in Statistics from Washington University in St. Louis, Sep 2022 - August 2025, first position: Assistant Professor at Kennesaw State University.
4. Dr. Salil Koner, PhD in Statistics from North Carolina State University, Feb 2022 - June 2025, first position: Assistant Professor at University of California, Riverside.
5. Dr. Xiaoxiao Zhou, PhD in Statistics from The Chinese University of Hong Kong, March 2023 - Feb 2024, first position: Assistant Professor at University of Alabama at Birmingham.
6. Dr. Xiaoming Xu, PhD in Statistics from Colorado State University, July 2021 - Nov 2023, first position: AbbVie.
7. Dr. Yuanyuan Guo, PhD in Statistics from University of Missouri, July 2021 - Nov 2022.
8. Dr. Xiaofang Yan, PhD in Biostatistics from University of Louisville, Jan 2020 - Nov 2020, first position: Medpace.
9. Dr. Cai Li, PhD in Statistics from North Carolina State University, Jan 2019 - Sep 2019, first position: Assistant Member at St. Jude Children's Research Hospital.

Classroom Teaching

Department of Biostatistics & Bioinformatics, Duke University

1. Categorical Data Analysis, Bio 714, Spring 2019, lead instructor, 100% responsibility.

Department of Biostatistics, University of Texas School of Public Health

1. Statistical Computing, PH 1998, Spring 2009, lead instructor, 100% responsibility.
2. Bayesian Data Analysis, PH 1998, Fall 2009, lead instructor, 100% responsibility.
3. Statistical Computing, PH 1998, Fall 2010, lead instructor, 100% responsibility.
4. Bayesian Data Analysis, PH 1998, Spring 2011, lead instructor, 100% responsibility.
5. Statistical Computing, PH 1998, Fall 2011, lead instructor, 100% responsibility.
6. Statistical Computing, PH 1930, Fall 2012, lead instructor, 100% responsibility.
7. Bayesian Data Analysis, PH 1965, Spring 2013, lead instructor, 100% responsibility.
8. Statistical Computing, PH 1930, Fall 2013, lead instructor, 100% responsibility.
9. Statistical Computing, PH 1930, Fall 2014, lead instructor, 100% responsibility.
10. Statistical Methods in Correlated Outcomes, PH 1918, Spring 2016, lead instructor, 100% responsibility.
11. Statistical Computing, PH 1930, Fall 2016, lead instructor, 100% responsibility.

Primary PhD dissertation advisor

1. Min Yi, doctoral in Biostatistics from UTHealth, graduated in Spring 2011, dissertation title: "Bayesian and survival model for tumor relapse status and disease-specific survival, with applications to breast cancer". First position: Senior Biostatistician at MD Anderson Cancer Center.
2. Geng Chen, doctoral in Biostatistics from UTHealth, graduated in Fall 2014, dissertation title: "Bayesian inference for multivariate longitudinal data analysis using robust distributions". First position: Senior biostatistician at GlaxoSmithKline.
3. Huirong Zhu, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 5, 2015, dissertation title: "Two-part mixture models for zero-inflated heterogeneous measurements and time to event data", co-advise with Stacia DeSantis, PhD. First position: Biostatistician at Texas Children Hospital.
4. Li-An Lin, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 19, 2015, dissertation title: "Bayesian analysis of multi-type recurrent events with dependent termination". First position: Senior biostatistician at Merck.
5. Jue Wang, doctoral in Biostatistics from UTHealth, dissertation defense occurred on May 19, 2016, dissertation title: "Dynamic predictions from joint models for multivariate longitudinal measurements and event time data". First position: Senior biostatistician at Genentech.
6. Ming Yang, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 29, 2016, dissertation title: "Bayesian quantile regression joint models: inference and dynamic predictions", co-advise with Stacia DeSantis, PhD. First position: Senior biostatistician at Genentech.
7. Kan Li, doctoral in Biostatistics from UTHealth, dissertation defense occurred on Feb 26, 2018, dissertation title: "Functional joint models for longitudinal and time-to-event data". First position: Senior biostatistician at Merck.
8. Jun Zhang, doctoral in Biostatistics from UTHealth, dissertation defense occurred on November 6, 2018, dissertation title: "Bayesian semi-parametric joint modeling for Parkinson's disease progression, competing failure times and missing data". First position: Senior biostatistician at Blue Cross Blue Shield.
9. Xuehan Ren, doctoral in Biostatistics from UTHealth, dissertation defense occurred on March 12, 2019, dissertation title: "Dynamic prediction of recurrent cardiovascular events". First position: Senior biostatistician at Gilead Sciences.
10. Jeffery Lin, doctoral study in Biostatistics, UTHealth, dissertation defense occurred on April 8, 2022, dissertation title: "Machine learning for the joint analysis of multivariate longitudinal and survival data". First position: Senior biostatistician at GlaxoSmithKline.
11. Haotian Zou, doctoral student in Biostatistics, The University of North Carolina at Chapel Hill, dissertation defense occurred on May 19, 2023, dissertation title: "Dynamic prediction with multivariate longitudinal outcomes and longitudinal magnetic resonance imaging data". First position: Postdoctoral fellow at Duke University.
12. Dongrak Choi, doctoral student in Biostatistics & Bioinformatics, Duke University, dissertation defense occurred on January 26, 2024, dissertation title: "An integrative modeling framework for neurodegenerative studies with longitudinal clinical data". First position: Senior biostatistician at Edwards Lifesciences Corp.

PhD dissertation committee

1. Lin Huo, doctoral in Biostatistics, UTHealth, graduated in Spring 2011, dissertation title: "Bayesian adaptive design for early phase drug-combination clinical trials".
2. Yunfei Wang, doctoral in Biostatistics, UTHealth, graduated in Spring 2012, dissertation title: "A simulation study using the hybrid statistics and the meta-analysis t-test for data with matched and unmatched subjects in the interim analysis of clinical trials".

3. Jin Young Byun, doctoral in Biostatistics, UTHealth, graduated in Spring 2012, dissertation title: "A hybrid method in combining treatment effects from matched and unmatched studies".
4. Rui Xia, doctoral in Biostatistics, UTHealth, graduated in year 2014, dissertation title: "A likelihood-based statistical method to detect DNA aberrations in SNP array data of tumor samples with very high normal cell contamination using haplotype structure derived from population genotypes".
5. Austin Brown, doctoral in Epidemiology, UTHealth, graduated in Spring 2014, dissertation title: "Healthcase professional weight loss advice, diet, physical activity and obesity in the fire service".
6. Julia Benoit, doctoral in Biostatistics, UTHealth, graduated in Spring 2014, dissertation title: "A hidden Markov model approach to analyze longitudinal ternary outcome subject to possible misclassification".
7. Michele Vidoni, doctoral student in Epidemiology, UTHealth, graduated in Spring 2016, dissertation title: "Vitamin B12 and homocysteine associations with physical function measures in older adults, data from the Baltimore longitudinal study of aging".
8. Yi-Ju Chiang, doctoral student in Biostatistics, UTHealth, dissertation proposal defense occurred on August 18, 2016, dissertation title: "Bivariate longitudinal logistic models with misclassified covariates and outcomes".

Primary MS thesis advisor

1. Eunji Jo, master student in Biostatistics, graduated in Fall 2009, thesis title: "Comparison of Bayesian and frequentist approaches in clinical outcomes between hospitals".
2. Minjeong Park, master student in Biostatistics, graduated in Spring 2010, thesis title: "Effects of air pollution on asthma hospitalization rates in different age groups in metropolitan cities of Korea".
3. Xiao Su, master in Biostatistics, graduated in May 2012, thesis title: "Bayesian estimation of multilevel item response model with missing data".
4. Gongbo Xie, master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Model for Predicting Efficacy of Treatment and Complications in Women after Intradetrusor OnabotulinumtoxinA for Non-Neurogenic Urgency Incontinence".
5. Chun Xu, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Physical function and accelerometer-based physical activity features in Physical Performance Across the Life-span Study (PALS)".
6. Huiyue Li, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2022, thesis title: "Novel gene variant in TP37, PI3R1, CALM1, and PLCG2 of the neurotrophin signaling pathway are associated with the progression from MCI to AD".
7. Caihan Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2022, thesis title: "Novel genetic variants in JAK2, CALM1 and PRKCA of the Biocarta biopeptide pathway are associated with progression from MCI to AD".
8. Songyun Zhang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2023, thesis title: "Assessing the necessity of regional nodal radiotherapy for breast cancer patients with positive ALN but pathologically regional lymph node negative disease after neoadjuvant chemotherapy".
9. Guanlin (Emery) Guo, Master Student in Duke Biostatistics & Bioinformatics, to graduate in Spring 2024, thesis title: "Explainable Graph-Neural Network on neuroimaging-based cortical surface data to identify people with risk of Alzheimer's Disease".

10. Linke Li, Master Student in Duke Biostatistics & Bioinformatics, to graduate in Spring 2024, thesis title: "biologically-explainable deep-learning based polygenic genetic risk score (PRS) for Alzheimer's disease".

MS students thesis committee

1. Meilin Jiang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Data Visualization for Intensive Self-monitoring Glucose Trajectories of Type II Diabetes Patients and Predictive Modeling for Hemoglobin A1c Values".
2. Zhecheng Sheng, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Analyzing the association of time-varying vital signs with in-hospital mortality & ICU transfer".
3. Meingdi Xuan, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Comparing survival analysis methods in estimating treatment effect when treatment switching in a randomized clinical trial".
4. Farzana Noor Bindu, Master Student in Biostatistics, graduated in Spring 2012, thesis title: "Performance of classifying diabetes and diagnostic threshold for diabetes among the Mexican American border community".
5. Jin Li, Master Student in Biostatistics, graduated in Fall 2011, thesis title: "Nonlinear dimension reduction in pathway-based genome-wide association analysis".
6. Huandong Sun, Master Student in Biostatistics, graduated in Spring 2012, thesis title: "A computerized statistical framework for coalescent analysis".
7. Meilin Jiang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Data Visualization for Intensive Self-monitoring Glucose Trajectories of Type II Diabetes Patients and Predictive Modeling for Hemoglobin A1c Values".
8. Zhecheng Sheng, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Analyzing the association of time-varying vital signs with in-hospital mortality & ICU transfer".
9. Mengdi Xuan, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2019, thesis title: "Comparing survival analysis methods in estimating treatment effect when treatment switching in a randomized clinical trial".
10. Siqi Zhang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Building a Prediction Model for Patient Enrollment in Clinical Trials Using Publicly Available Databases".
11. Chenyang Wang, Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2020, thesis title: "Bayesian Network Meta-Analysis of Trials Comparing Treatments for Schizophrenia".
12. Yijie He, 2nd year Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2021, thesis title: "Race Adjusted DNA damage repair SNP Analysis of C90401 Cohort for Prostate Cancer Association".
13. Blessing Ibe, 2nd year Master Student in Duke Biostatistics & Bioinformatics, graduated in Spring 2022, thesis title: "Bayesian individual patient-level network meta-analysis of trials Comparing treatments for bipolar disorder".

MPH students committee

1. Naalia Sami, MPH in Biostatistics, graduated in Spring 2013, thesis title: "Effect of insurance status on stage of breast cancer diagnosis".
2. Simit Doshi, MPH graduate, thesis title: "Potential of role of statins as immunomodulators in treating pneumococcal pneumonia".

3. Jamie Marie Emert, MPH graduate, thesis title: "SEIS Community Needs Assessment Study for the Motherland and Hour Clinics in Houston, Texas".
4. Ashley A Francis, MPH graduate, thesis title: "Health significance of chlorination byproducts in drinking water: the Houston experience: a continuation study".
5. Oscar Sean Friendly, MPH graduate, thesis title: "Characterization of the east Houston particulate matter < 2.5 atmosphere: a novel method for desorbing organic constituents from airborne particulates".
6. Jacky T Lee, MPH graduate.
7. Alexander Charles Reese, MPH graduate, thesis title: "The effect of CATCH school lunch programs on obesity: a school based secondary analysis cross sectional study".
8. Yanneth Rivera, MPH graduate.

Invited Talks, Presentations, and Posters

Invited talks

1. Analysis of Smoking Cessation Patterns Using a Stochastic Mixed Effects Model with a Latent Cured State, *National Cancer Institute*, June, 2007.
2. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Department of Biostatistics, University of Texas at Houston*, November 2007.
3. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Department of Biostatistics, The University of North Carolina at Chapel Hill*, January 2008.
4. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Division of Biostatistics, Thomas Jefferson University*, January 2008.
5. Mixed Effects Stochastic Process Models of Smoking Cessation Behavior, *Department of Biostatistics, University of Florida*, February 2008.
6. Functional and Dynamic Data Analysis to Biomedical Research - Topic Contributed, *Biometrics Section, Joint Statistical Meetings (JSM)*, August 2009, Washington DC, session number: 254.
7. Classifying Ipsilateral Breast Tumor Relapse: A Novel Bayesian Approach, *Department of Biostatistics, University of Texas at Houston*, September 2009.
8. Binomial Regression with a Misclassified Covariate and Outcome, *Department of Biostatistics, University of Texas at Houston*, April 2010.
9. Center for AIDS Research and Analysis Core: Research Opportunities, *Baylor College of Medicine*, November 2010.
10. Methodological issues of using Item Response Models in Parkinson Disease Studies, *Department of Biostatistics, University of Texas at Houston*, April 2012.
11. Recent development of Multilevel Item Response Models with applications to Parkinson's Disease, *Department of Biostatistics, University of Texas at Houston*, January 2013.
12. The Use of Multilevel Item Response Models in Studying Parkinson's Disease, *Department of Mathematics, University of Texas at Arlington*, March 2013.
13. Robust Bayesian Inference for Multivariate Longitudinal Data Using Normal/Independent Distributions, *Houston Area Chapter, American Statistical Association*, February 2014.
14. Joint model of multivariate longitudinal data using multilevel item response model with applications to Parkinson's Disease, *The Western North American Region of the International Biometric Society Meeting (WNAR)*, June 2014, Honolulu, session number: WNAR Invited 1.
15. Bayesian Hierarchical Model for Multiple Repeated Measures and Survival Data: an application to Parkinson's Disease, *Third Joint Biostatistics Symposium*, June 2014, Chengdu, China.

16. Topic contributed: Joint Analysis of Multivariate Longitudinal Ordinal Measurements and Survival Data: An Application to Parkinson's Disease, *JSM*, August 2014, Boston, session number: 326.
17. Bayesian Analysis of Multi-type Recurrent Events and Dependent Termination with Nonparametric Covariate Functions, *Department of Preventive Medicine-Biostatistics, Northwestern University Feinberg School of Medicine*, October 6, 2014.
18. Bayesian Analysis of Multi-type Recurrent Events and Dependent Termination with Nonparametric Covariate Functions, *Department of Biostatistics & Epidemiology, University of Pennsylvania Perelman School of Medicine*, November 20, 2014.
19. Effect of Creatine Monohydrate on Clinical Progression in Patients With Parkinson Disease: A Randomized Clinical Trial, *Department of Biostatistics, University of Texas at Houston*, April 17, 2015.
20. Simultaneous inference of a misclassified outcome and competing risks failure time data, *Joint 24th ICSA Applied Statistics Symposium and 13th Graybill Conference*, June 2015, Fort Collins, Colorado, session number: 13.
21. Dynamic Prediction of Alzheimer's Disease Risk Based on Longitudinal Biomarkers and Functional Data, *JSM*, August 2016, Chicago, session number: 341.
22. Dynamic Prediction for Multiple Repeated Measures and Event Time Data: an Application to Parkinson's Disease, *The 10th ICSA International Conference*, December 2016, Shanghai, China, session number: 92.
23. Dynamic Prediction of Alzheimer's Disease Progression with Longitudinal Functional Joint Model, *2017 Confence on Lifetime Data Science*, May 2017, Storrs, Connecticut, session number: 10.
24. Functional Joint Model with application of Alzheimer's disease, *2017 ICSA Applied Statistics Symposium*, June 2017, Chicago, session number: 78.
25. Dynamic Risk Prediction and Assessment, *Precision Medicine: Statistical Challenges and Opportunities Conference*, June 2017, Chicago, session number: 78.
26. Bayesian Functional Joint Models for Longitudinal and Time-To-Event Data, Department of Statistics, North Carolina State University, October 12, 2017.
27. Dynamic Risk Prediction and Assessment, Precision Medicine: Statistical Challenges and Opportunities Conference, May 8, 2018.
28. Dynamic Prediction of Alzheimer's Disease Progression: the Big Data Approach, Duke Clinical Research Institute, April 24, 2018.
29. Dynamic Risk Assessment and Prediction, *ASA/AdvaMed Medical Device Statistical Issues Conference*, May 8, 2018, Washington, DC.
30. Dynamic prediction using multivariate functional joint model: an application to Alzheimer's disease, *2018 International Chinese Statistical Association China Conference*, July 2018, Qingdao, China, session number: 6B.
31. Missing Data Issues in the Studies of Neurodegenerative Disorders: The Methodology, *JSM*, August 2018, Vancouver, session number: 44.
32. Dynamic Prediction of Alzheimer's Disease Progression Using Features of Multiple Longitudinal Outcomes and Time-to-Event Data, *The 2nd Conference on Lifetime Data Science*, May 2019, University of Pittsburgh, session number: 13.
33. Precision Medicine for Alzheimer's Disease Monitoring and Progression: the Dynamic Prediction Approach, *Division of Biostatistics, Washington University in St. Louis*, October 4, 2019.
34. Integrative Modeling and Dynamic Prediction of Alzheimer's Disease, *The Eastern North American Region of the International Biometric Society Spring Meeting (ENAR)*, Mar 2020, virtual meeting due to COVID-19, session number: 33.
35. Functional Data Analysis: Novel Statistical Methods and Applications in Medical Research, *Data-Driven Precision Medicine and Transnational Research in the Era of Big Data*, St. Jude Children Research Hospital, May 16, 2020.
36. Functional Data Analysis: Novel Statistical Methods and Applications in Alzheimer' Disease Research, *University of Nevada, Las Vegas*, April 28, 2022.

37. Functional Data Analysis: Novel Statistical Methods and Applications in Alzheimer' Disease Research, *The University of Mississippi Medical Center, Jackson, MS*, Sep 8, 2022.
38. Dynamic Prediction with Multivariate Longitudinal Outcomes and Longitudinal Magnetic Resonance Imaging Data, *ENAR*, Mar 2023, Nashville, session number: 13.
39. Deep Learning for the Dynamic Prediction of Multivariate Longitudinal and Survival Data, *JSM*, August 2023, Toronto, session number: 1172.
40. Integrative Modeling and Dynamic Prediction of Alzheimer's Disease, *University of Pittsburgh, Department of Biostatistics, Pittsburgh, PA*, Sep 7, 2023.
41. Integrative Modeling and Dynamic Prediction of Alzheimer's Disease, *North Carolina State University, Department of Statistics, Raleigh, NC*, Jan 16, 2024.
42. Advancing Alzheimer's Drug Development: A Novel Longitudinal Rank-Sum Test for Multiple Primary Endpoints in Randomized Controlled Trials, *ENAR*, Mar 2024, Baltimore, session number: 43.

Presentations and posters

1. Contributed Talk: Modeling Individual Addiction Behavior Using a Mixed-Effect Model with Three States, *Joint Statistical Meetings (JSM)*, August 2006, Seattle, session number: 199.
2. Contributed Talk: Analysis of Smoking Cessation Patterns Using a Stochastic Mixed Effects Model with a Latent Cured State, *The Eastern North American Region of the International Biometric Society Spring Meeting (ENAR)*, March 2007, Atlanta, session number: 83.
3. Contributed Talk: Analysis of Smoking Cessation Patterns Using a Stochastic Mixed Effects Model with a Latent Cured State, *JSM*, August 2007, Salt Lake City, session number: 426.
4. Contributed Talk: Shrinkage Estimation for Robust and Efficient Screening of single SNP Association from Case-Control Genome-wide Association Studies, *ENAR*, March 2008, Arlington, session number: 45.
5. Contributed Talk: A Bayesian Approach for Correcting Misclassification in both Outcome Variable and Covariate, *ENAR*, March 2010, New Orleans, session number: 34.
6. Contributed Talk: Multivariate Longitudinal Bayesian Imputation for Global Statistical Test, with Applications to Parkinson's Disease, *JSM*, August 2010, Vancouver, Canada, session number: 128.
7. Contributed Talk: A Hybrid Method in Combining Treatment Effects from Matched and Unmatched Studies, presented by Jinyoung Byun*, *JSM*, August 2010, Vancouver, Canada, session number: 631.
8. Contributed Talk: A Bayesian Model for Misclassified Binary Outcomes and Correlated Survival Data, with Applications to Breast Cancer, presented by Min Yi*, *JSM*, August 2010, Vancouver, Canada, session number: 163.
9. Contributed Talk: Robust Bayesian Inference for Multivariate Longitudinal Data with Normal/Independent Distributions, *ENAR*, April 2012, Washington DC, session number: 78.
10. Poster Presentation: Analysis of Parkinson's Disease Data Using Item Response Model and Normal/Independent Distributions, *The International Parkinson and Movement Disorder Society's 16th International Congress*, June 2012, Dublin, Ireland, abstract number: 1297.
11. Contributed Talk: A Bayesian Item Response Model for Nonignorable Missing Data in Parkinson Disease Patients, *JSM*, July 2012, San Diego, session number: 244.
12. Contributed Talk: A Bayesian Approach to Joint Analysis of Parametric Accelerated Failure Time and Multivariate Longitudinal Data, *ENAR*, March 2013, Orlando, session number: 106.
13. Poster Presentation: Bayesian multiple imputation for missing multivariate longitudinal data from a Parkinson's disease clinical trial: a Power Analysis, *Society for Clinical Trials' 34th Annual Meeting*, May 2013, Boston.
14. Poster Presentation: Methods for Imputing Missing Data for Parkinson's Disease Clinical Trials with Multiple Correlated Outcomes, *The International Parkinson and Movement Disorder Society's 17th International Congress*, June 2013, Sydney, Australia, abstract number: 478.

15. Contributed Talk: Joint Modeling of Multivariate Longitudinal Measurements and Survival Data with Applications to Parkinson's Disease, *JSM*, August 2013, Montreal, Canada, session number: 308.
16. Session Chair: Developments in Genetic Association Studies, *JSM*, August 2013, Montreal, Canada, session number: 537.
17. Poster Presentation: Methods for Imputing Missing Data for Parkinson's Disease Clinical Trials with Multiple Correlated Outcomes, *Texas Regional Clinical and Translational Science Award (CTSA) Consortium Meeting*, November 2013, San Antonio.
18. Contributed Talk: Joint model for a diagnostic test without a gold standard in the presence of a dependent terminal event, *ENAR*, March 2014, Baltimore, session number: 72.
19. Session Chair: Imaging, *ENAR*, March 2014, Baltimore, session number: 58.
20. Session Chair: Recent Advances in Missing Data Methods, *The Western North American Region of the International Biometric Society Meeting (WNAR)*, June 2014, Honolulu.
21. Session Chair: Methods for Current-Status and Interval-Censored Data, *JSM*, August 2015, Seattle, session number: 80.
22. Contributed Talk: Dynamic Predictions from Joint Models for Multivariate Longitudinal Measurements and Survival Data, *JSM*, August 2015, Seattle, session number: 608.
23. Poster Presentation: Dynamic prediction for multiple repeated measures and event time data: An application to Parkinson's disease, *The International Parkinson and Movement Disorder Society's 20th International Congress*, June 2016, Berlin, Germany, abstract number: 1869.
24. Contributed Talk: Dynamic Prediction of Alzheimer's Disease Risk Based on Longitudinal Biomarkers and Functional Data, *The 4th IMS Asian Pacific Rim Meeting*, June 2016, Hong Kong, session number: CP19.
25. Session Chair: Recent Challenges in -Omics Sciences, *JSM*, August 2016, Chicago, session number: 428.
26. Poster Presentation: Bayesian Analysis of Multiple Longitudinal Outcomes of Mixed Types with Nonparametric Treatment Effects, *ENAR*, March 2017, Washington DC.
27. Poster Presentation: Differential Item Functioning in the Unified Dyskinesia Rating Scale (UDysRS), *The International Parkinson and Movement Disorder Society's 21th International Congress*, June 2017, Vancouver, Canada, abstract number: 1141.
28. Poster Presentation: Missing Data in the Unified Dyskinesia Rating Scale (UDysRS), *The International Parkinson and Movement Disorder Society's 22nd International Congress*, Oct 2018, Hong Kong, China, abstract number: 1797.
29. Session Chair: ASA Biometrics Section JSM Travel Awards (I), *JSM*, July 2019, Denver, session number: 393.
30. Poster Presentation: Successful use of UDysRS regardless of PD- or Dyskinesia-duration, *The International Parkinson and Movement Disorder Society's 23rd International Congress*, Sep 2019, Nice, France, abstract number: 1190.